

Complete Cisco Packet Tracer Network Configuration Guide

ICS 605 – Network Security Project (Milestone 1)

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Course: Network Security (Spring 2026)

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18. Quick Verification Checklist

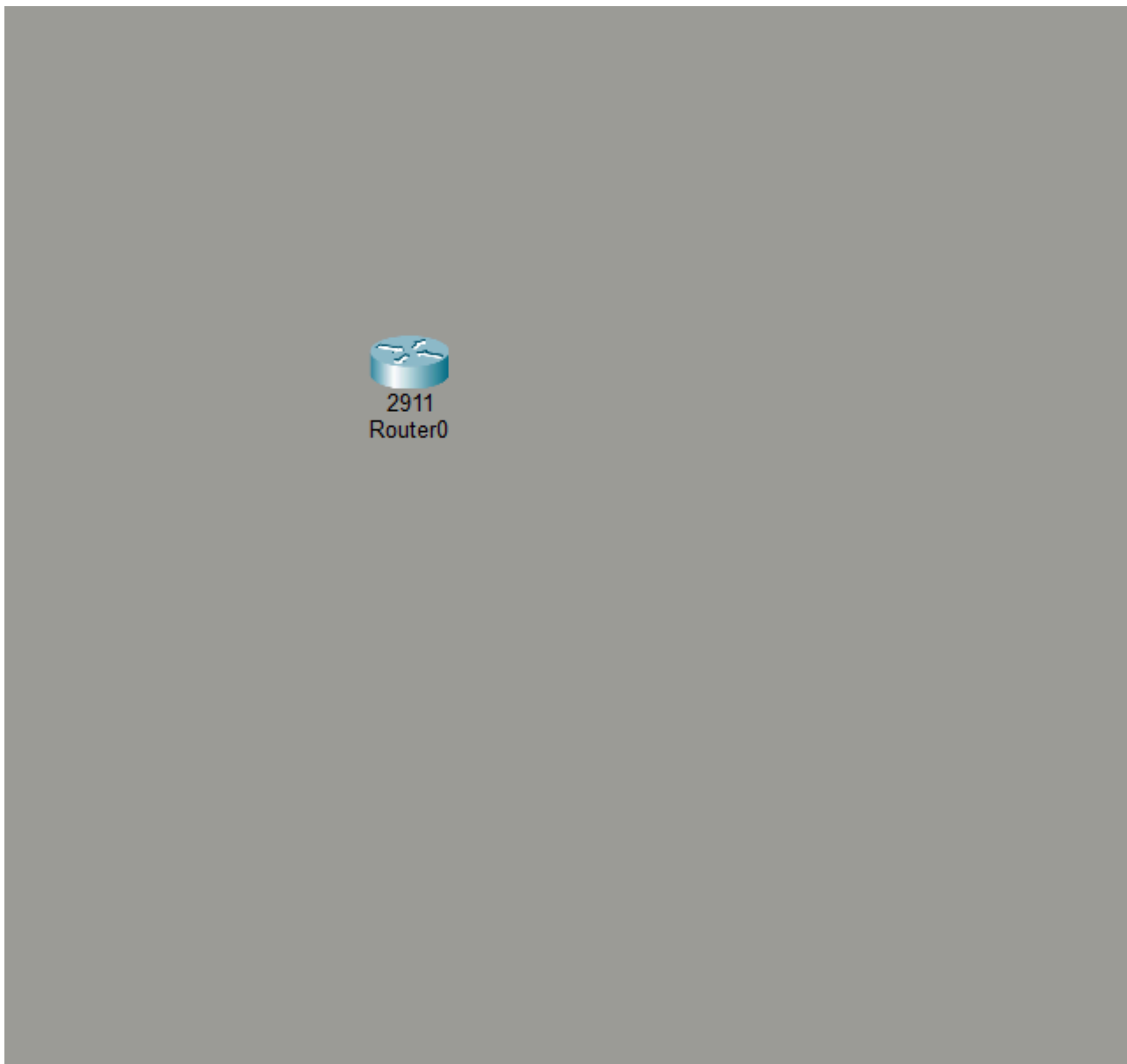
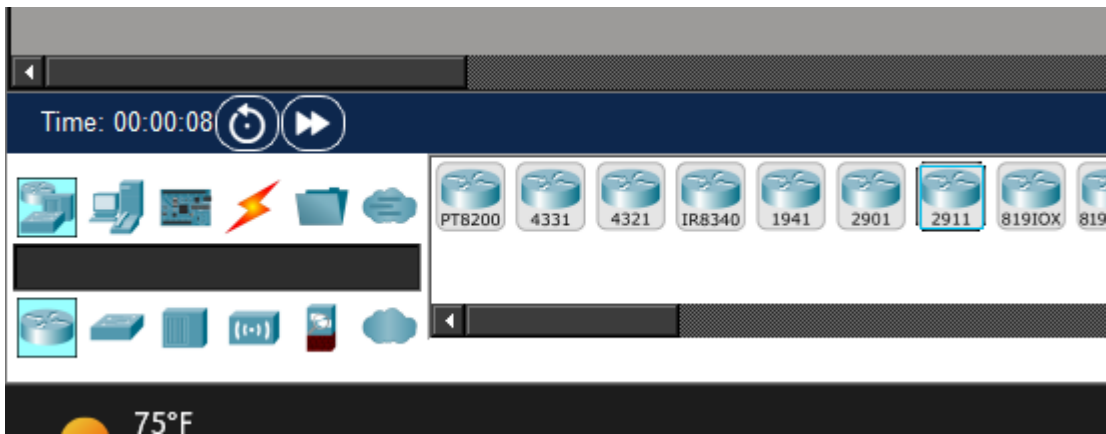
1. Packet Tracer Basics for New Users

Launching Packet Tracer

- Open Cisco Packet Tracer. You will see a blank canvas (Workspace).
- **Bottom-left panel:** Device categories (Routers, Switches, End Devices, etc.).
- **Bottom-right panel:** Specific device models within that category.
- **Drag** a device from the bottom panel onto the canvas to place it.

Placing Devices

1. Click a **device category** in the bottom-left (e.g., "Routers").
2. Click a **specific model** in the bottom-right (e.g., "2911").
3. Click anywhere on the canvas to place it. Click again to place another. Press **Esc** to stop placing.



Connecting Devices (Cables)

1. Click the **lightning bolt icon** (Connections) in the bottom-left panel.
2. Select a cable type from the bottom-right:
 - **Copper Straight-Through** (solid orange line) – PC/Server to Switch, Router to Switch
 - **Copper Cross-Over** (dashed orange line) – Switch to Switch, PC to PC (rarely needed with auto-MDI)
 - **Console** (light blue line) – Laptop/PC RS-232 to Router/Switch Console port

- **Serial DCE** (red line with clock symbol) – Router Serial to Router Serial (one end is DCE)
3. Click the **source device**, then select the **source port** from the pop-up.
 4. Click the **destination device**, then select the **destination port**.
 5. Green dots on cable ends = link up. Red dots = link down (normal until configured).

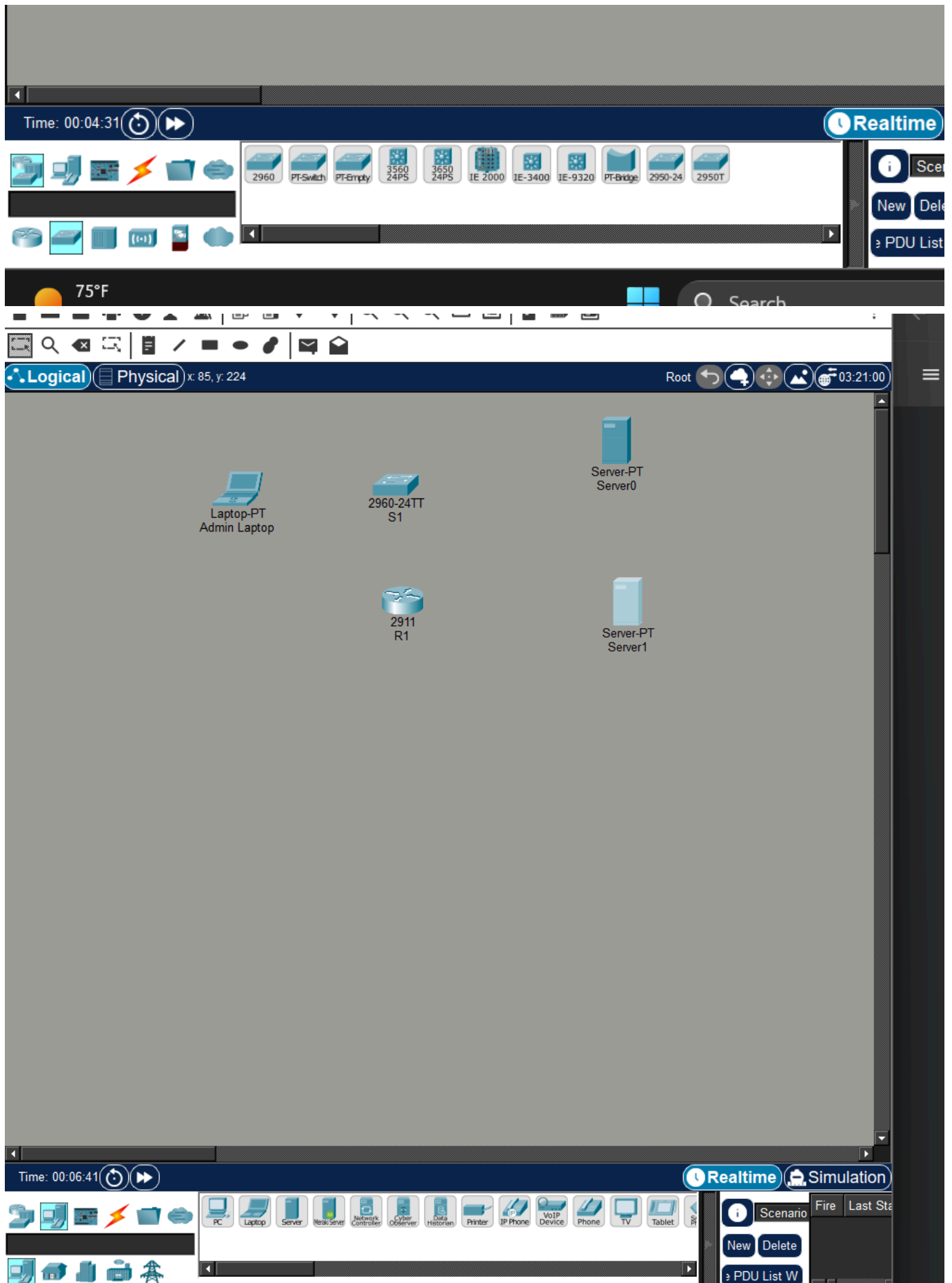
Accessing Device CLI

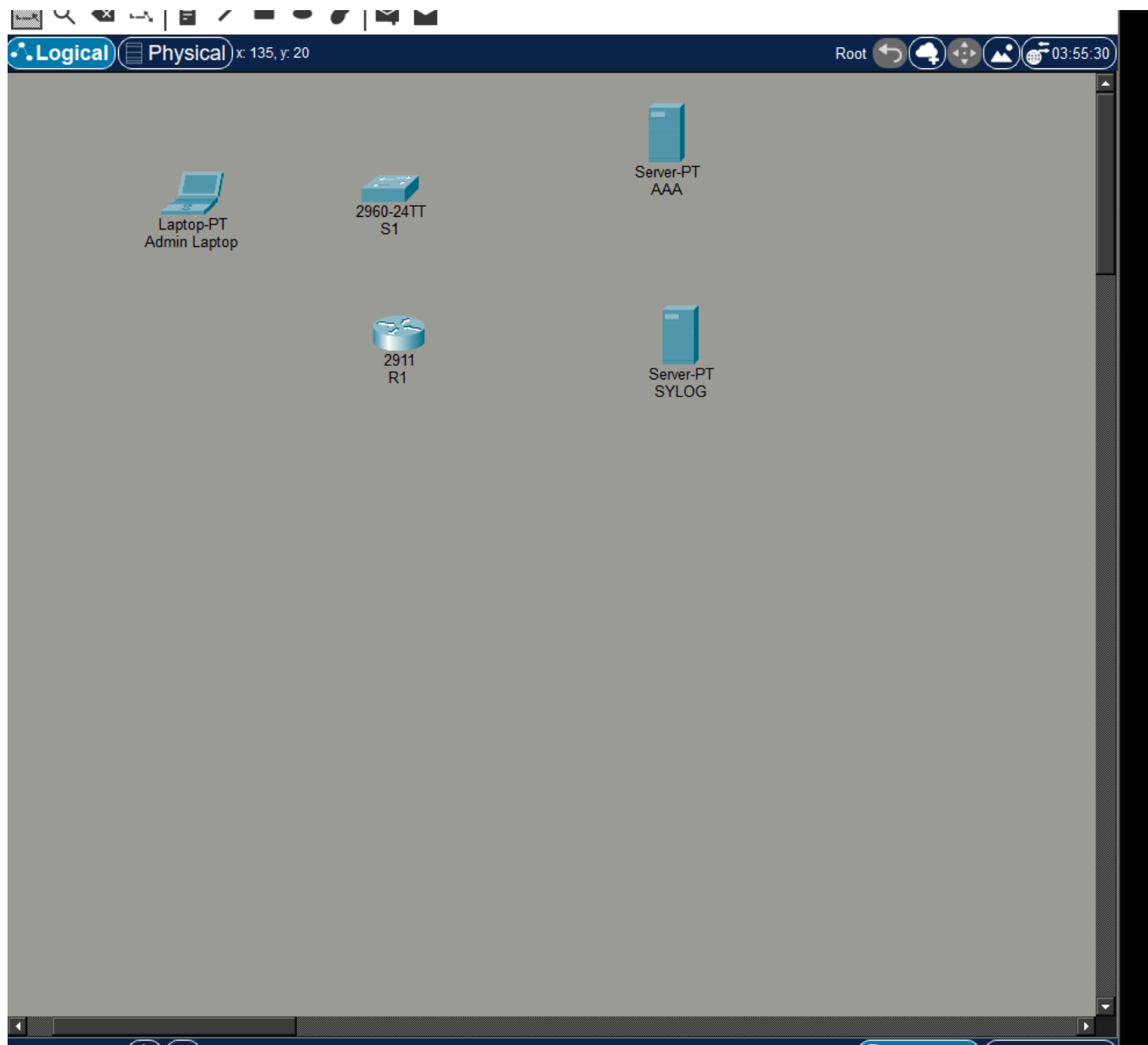
- **Double-click** any router or switch → Click the **CLI** tab → Press **Enter** to start.
- For a laptop/PC → Double-click → **Desktop** tab → **Terminal** (for console) or **Command Prompt** (for ping/tests).

Adding Hardware Modules to Routers (CRITICAL for Serial Ports)

The Cisco 2911 router does **not** have Serial ports by default in Packet Tracer. You must add them manually:

1. Double-click the router → Click the **Physical** tab.
2. Click the **power button** (green button on the router image) to **turn the router OFF**.
3. In the left panel, find "**HWIC-2T**" (scroll down the module list).
4. **Drag** the HWIC-2T module to an empty slot on the router chassis image (usually visible as an empty rectangular slot).
5. Click the **power button** again to turn the router **back ON**.
6. The router now has **Serial0/0/0** and **Serial0/0/1** interfaces.





Do this for R1 and R2 before making any serial connections. R3 only needs one serial port so one HWIC-2T is sufficient.

Naming Convention Used in This Guide

The topology explicitly names devices R1, R2, R3 and S1–S4. When Packet Tracer places a router, it defaults to "Router0", "Router1", etc. We will rename all devices using the `hostname` command in the CLI.

2. Network Overview & IP Addressing Table

Subnets Summary

Subnet	Purpose	Network	Gateway
Server Room	Internal servers + admin	192.168.1.0/24	192.168.1.1 (R1 Gi0/0)
Tech Support	IT employee PCs	192.168.2.0/24	192.168.2.1 (R1 Gi0/1)
Employees	Sales, Marketing, Meeting, Mgmt	192.168.3.0/24	192.168.3.1 (R1 Gi0/2)

Subnet	Purpose	Network	Gateway
R1–R2 WAN Link	Internal backbone	172.16.1.0/24	—
DMZ	Public-facing servers	172.16.3.0/24	172.16.3.1 (R2 Gi0/0)
R2–R3 WAN Link	ISP uplink	172.16.2.0/24	—
Internet	Simulated external network	209.165.200.0/24	209.165.200.1 (R3 Gi0/0)

Complete IP Address Table

Device	Label in PT	Interface	IP Address	Subnet Mask	Default Gateway
Router R1	R1	GigabitEthernet0/0	192.168.1.1	255.255.255.0	—
Router R1	R1	GigabitEthernet0/1	192.168.2.1	255.255.255.0	—
Router R1	R1	GigabitEthernet0/2	192.168.3.1	255.255.255.0	—
Router R1	R1	Serial0/0/0	172.16.1.1	255.255.255.0	—
Router R2	R2	Serial0/0/0	172.16.1.2	255.255.255.0	—
Router R2	R2	GigabitEthernet0/0	172.16.3.1	255.255.255.0	—
Router R2	R2	Serial0/0/1	172.16.2.1	255.255.255.0	—
Router R3	R3	Serial0/0/0	172.16.2.2	255.255.255.0	—
Router R3	R3	GigabitEthernet0/0	209.165.200.1	255.255.255.0	—
AAA Server	AAA-Server	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
SYSLOG Server	SYSLOG-Server	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
Admin Laptop	Admin-Laptop	FastEthernet0	192.168.1.4	255.255.255.0	192.168.1.1
IT PC 1	IT-PC1	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
IT PC 2	IT-PC2	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1
IT PC 3	IT-PC3	FastEthernet0	192.168.2.4	255.255.255.0	192.168.2.1
Sales PC 1	Sales-PC1	FastEthernet0	192.168.3.2	255.255.255.0	192.168.3.1
Sales PC 2	Sales-PC2	FastEthernet0	192.168.3.3	255.255.255.0	192.168.3.1
Sales PC 3	Sales-PC3	FastEthernet0	192.168.3.4	255.255.255.0	192.168.3.1
Marketing PC 1	Mktg-PC1	FastEthernet0	192.168.3.5	255.255.255.0	192.168.3.1
Marketing PC 2	Mktg-PC2	FastEthernet0	192.168.3.6	255.255.255.0	192.168.3.1
Marketing PC 3	Mktg-PC3	FastEthernet0	192.168.3.7	255.255.255.0	192.168.3.1

Device	Label in PT	Interface	IP Address	Subnet Mask	Default Gateway
Meeting Room PC	Meeting-PC	FastEthernet0	192.168.3.8	255.255.255.0	192.168.3.1
Management PC	Mgmt-PC	FastEthernet0	192.168.3.9	255.255.255.0	192.168.3.1
Web Server	Web-Server	FastEthernet0	172.16.3.2	255.255.255.0	172.16.3.1
FTP Server	FTP-Server	FastEthernet0	172.16.3.3	255.255.255.0	172.16.3.1
Email Server	Email-Server	FastEthernet0	172.16.3.4	255.255.255.0	172.16.3.1
Internet Server	Internet-Server	FastEthernet0	209.165.200.2	255.255.255.0	209.165.200.1

3. Complete Physical Connection Table

Cable Types: ST = Copper Straight-Through | Console = Rollover/Console (light blue) | DCE = Serial DCE

#	From Device	From Port	To Device	To Port	Cable Type
1	Admin-Laptop	RS-232	R1	Console	Console (Rollover)
2	Admin-Laptop	FastEthernet0	S1	FastEthernet0/4	Copper Straight-Through
3	R1	GigabitEthernet0/0	S1	GigabitEthernet0/1	Copper Straight-Through
4	S1	FastEthernet0/1	AAA-Server	FastEthernet0	Copper Straight-Through
5	S1	FastEthernet0/2	SYSLOG-Server	FastEthernet0	Copper Straight-Through
6	R1	GigabitEthernet0/1	S2	GigabitEthernet0/1	Copper Straight-Through
7	S2	FastEthernet0/1	IT-PC1	FastEthernet0	Copper Straight-Through
8	S2	FastEthernet0/2	IT-PC2	FastEthernet0	Copper Straight-Through
9	S2	FastEthernet0/3	IT-PC3	FastEthernet0	Copper Straight-Through

#	From Device	From Port	To Device	To Port	Cable Type
10	R1	GigabitEthernet0/2	S3	GigabitEthernet0/1	Copper Straight-Through
11	S3	FastEthernet0/1	Sales-PC1	FastEthernet0	Copper Straight-Through
12	S3	FastEthernet0/2	Sales-PC2	FastEthernet0	Copper Straight-Through
13	S3	FastEthernet0/3	Sales-PC3	FastEthernet0	Copper Straight-Through
14	S3	FastEthernet0/4	Mktg-PC1	FastEthernet0	Copper Straight-Through
15	S3	FastEthernet0/5	Mktg-PC2	FastEthernet0	Copper Straight-Through
16	S3	FastEthernet0/6	Mktg-PC3	FastEthernet0	Copper Straight-Through
17	S3	FastEthernet0/7	Meeting-PC	FastEthernet0	Copper Straight-Through
18	S3	FastEthernet0/8	Mgmt-PC	FastEthernet0	Copper Straight-Through
19	R1	Serial0/0/0 (DCE)	R2	Serial0/0/0 (DTE)	Serial DCE
20	R2	GigabitEthernet0/0	S4	GigabitEthernet0/1	Copper Straight-Through
21	S4	FastEthernet0/1	Web-Server	FastEthernet0	Copper Straight-Through
22	S4	FastEthernet0/2	FTP-Server	FastEthernet0	Copper Straight-Through
23	S4	FastEthernet0/3	Email-Server	FastEthernet0	Copper Straight-Through
24	R2	Serial0/0/1 (DCE)	R3	Serial0/0/0 (DTE)	Serial DCE
25	R3	GigabitEthernet0/0	S5*	GigabitEthernet0/1	Copper Straight-Through
26	S5*	FastEthernet0/1	Internet-Server	FastEthernet0	Copper Straight-Through

*S5 is an optional extra switch for the internet simulation. You may connect R3 Gi0/0 directly to Internet-Server FastEthernet0 using a Copper Straight-Through cable if you prefer (no switch needed for the internet segment).

Serial DCE Note: When connecting with a Serial DCE cable in Packet Tracer, one end of the cable has a clock symbol — that is the DCE end. The device that receives the DCE end **provides** the clock rate. In this design, R1 is DCE for the R1→R2 link, and R2 is DCE for the R2→R3 link. The DCE router must issue the **clock rate 64000** command on its serial interface.

4. Part A – Building the Physical Topology

Step A1 – Place All Devices on the Canvas

Organize the canvas into logical zones (server room top-left, employees bottom-left, DMZ top-right, external top-far-right):

Routers (category: Routers → model: 2911):

- Place **3 routers**: R1, R2, R3

Switches (category: Switches → model: 2960-24TT):

- Place **4 switches**: S1, S2, S3, S4

Servers (category: End Devices → model: Server-PT):

- Place **7 servers**: AAA-Server, SYSLOG-Server, Web-Server, FTP-Server, Email-Server, Internet-Server

PCs (category: End Devices → model: PC-PT):

- Place **11 PCs**: IT-PC1, IT-PC2, IT-PC3, Sales-PC1, Sales-PC2, Sales-PC3, Mktg-PC1, Mktg-PC2, Mktg-PC3, Meeting-PC, Mgmt-PC

Laptop (category: End Devices → model: Laptop-PT):

- Place **1 laptop**: Admin-Laptop

To label a device: Double-click the device label that appears below it on the canvas and type the new name.

Step A2 – Add Serial Modules to R1, R2, and R3

For R1:

- Double-click R1 → Click **Physical** tab.
- Click the green **power button** on the router image to power it OFF (the fan animation stops).
- In the module list on the left, scroll to find **HWIC-2T**.
- Click and drag **HWIC-2T** to the **first empty module slot** on the router (the small rectangular slot on the router chassis image, usually next to the fixed ports).
- Click the power button again to power R1 **ON**.

For R2:

- Double-click R2 → Click **Physical** tab.
- Power **OFF** R2.

3. Drag **HWIC-2T** to the first empty slot on R2.
4. Power **ON** R2.

For R3:

1. Double-click R3 → Click **Physical** tab.
 2. Power **OFF** R3.
 3. Drag **HWIC-2T** to the first empty slot on R3.
 4. Power **ON** R3.
-

Step A3 – Make All Physical Connections

Click the **lightning bolt (Connections)** icon in the bottom-left panel. Make every connection listed below in order.

SERVER ROOM connections:**Connection 1 – Admin-Laptop Console to R1**

- Cable: **Console** (light blue)
- Click **Admin-Laptop** → select **RS-232**
- Click **R1** → select **Console**

Connection 2 – Admin-Laptop Ethernet to S1

- Cable: **Copper Straight-Through**
- Click **Admin-Laptop** → select **FastEthernet0**
- Click **S1** → select **FastEthernet0/4**

Connection 3 – R1 to S1 (Server Room uplink)

- Cable: **Copper Straight-Through**
- Click **R1** → select **GigabitEthernet0/0**
- Click **S1** → select **GigabitEthernet0/1**

Connection 4 – S1 to AAA-Server

- Cable: **Copper Straight-Through**
- Click **S1** → select **FastEthernet0/1**
- Click **AAA-Server** → select **FastEthernet0**

Connection 5 – S1 to SYSLOG-Server

- Cable: **Copper Straight-Through**
 - Click **S1** → select **FastEthernet0/2**
 - Click **SYSLOG-Server** → select **FastEthernet0**
-

TECHNICAL SUPPORT connections:

Connection 6 – R1 to S2 (Tech Support uplink)

- Cable: **Copper Straight-Through**
- Click **R1** → select **GigabitEthernet0/1**
- Click **S2** → select **GigabitEthernet0/1**

Connection 7 – S2 to IT-PC1

- Cable: **Copper Straight-Through**
- Click **S2** → select **FastEthernet0/1**
- Click **IT-PC1** → select **FastEthernet0**

Connection 8 – S2 to IT-PC2

- Cable: **Copper Straight-Through**
- Click **S2** → select **FastEthernet0/2**
- Click **IT-PC2** → select **FastEthernet0**

Connection 9 – S2 to IT-PC3

- Cable: **Copper Straight-Through**
 - Click **S2** → select **FastEthernet0/3**
 - Click **IT-PC3** → select **FastEthernet0**
-

EMPLOYEES' OFFICES connections:**Connection 10 – R1 to S3 (Employees uplink)**

- Cable: **Copper Straight-Through**
- Click **R1** → select **GigabitEthernet0/2**
- Click **S3** → select **GigabitEthernet0/1**

Connection 11 – S3 to Sales-PC1 (Sales Office)

- Cable: **Copper Straight-Through**
- Click **S3** → select **FastEthernet0/1**
- Click **Sales-PC1** → select **FastEthernet0**

Connection 12 – S3 to Sales-PC2 (Sales Office)

- Cable: **Copper Straight-Through**
- Click **S3** → select **FastEthernet0/2**
- Click **Sales-PC2** → select **FastEthernet0**

Connection 13 – S3 to Sales-PC3 (Sales Office)

- Cable: **Copper Straight-Through**
- Click **S3** → select **FastEthernet0/3**
- Click **Sales-PC3** → select **FastEthernet0**

Connection 14 – S3 to Mktg-PC1 (Marketing Office)

- Cable: **Copper Straight-Through**
- Click **S3** → select **FastEthernet0/4**
- Click **Mktg-PC1** → select **FastEthernet0**

Connection 15 – S3 to Mktg-PC2 (Marketing Office)

- Cable: **Copper Straight-Through**
- Click **S3** → select **FastEthernet0/5**
- Click **Mktg-PC2** → select **FastEthernet0**

Connection 16 – S3 to Mktg-PC3 (Marketing Office)

- Cable: **Copper Straight-Through**
- Click **S3** → select **FastEthernet0/6**
- Click **Mktg-PC3** → select **FastEthernet0**

Connection 17 – S3 to Meeting-PC (Meeting Room)

- Cable: **Copper Straight-Through**
- Click **S3** → select **FastEthernet0/7**
- Click **Meeting-PC** → select **FastEthernet0**

Connection 18 – S3 to Mgmt-PC (Management Office)

- Cable: **Copper Straight-Through**
- Click **S3** → select **FastEthernet0/8**
- Click **Mgmt-PC** → select **FastEthernet0**

WAN LINK – R1 to R2 (Internal ↔ DMZ):

Connection 19 – R1 (DCE) to R2 (DTE) via Serial

- Cable: **Serial DCE**
- Click **R1** → select **Serial0/0/0** (make sure the DCE clock icon end connects here)
- Click **R2** → select **Serial0/0/0**

After placing the cable, hover over the cable on R1's side — you should see a small clock icon indicating R1 is the DCE end. If the clock is on R2's side, note that R2 would then be DCE; in that case, swap which router gets **clock rate** in the configuration below.

DMZ connections:

Connection 20 – R2 to S4 (DMZ uplink)

- Cable: **Copper Straight-Through**
- Click **R2** → select **GigabitEthernet0/0**
- Click **S4** → select **GigabitEthernet0/1**

Connection 21 – S4 to Web-Server

- Cable: **Copper Straight-Through**
- Click **S4** → select **FastEthernet0/1**
- Click **Web-Server** → select **FastEthernet0**

Connection 22 – S4 to FTP-Server

- Cable: **Copper Straight-Through**
- Click **S4** → select **FastEthernet0/2**
- Click **FTP-Server** → select **FastEthernet0**

Connection 23 – S4 to Email-Server

- Cable: **Copper Straight-Through**
- Click **S4** → select **FastEthernet0/3**
- Click **Email-Server** → select **FastEthernet0**

WAN LINK – R2 to R3 (DMZ ↔ ISP):

Connection 24 – R2 (DCE) to R3 (DTE) via Serial

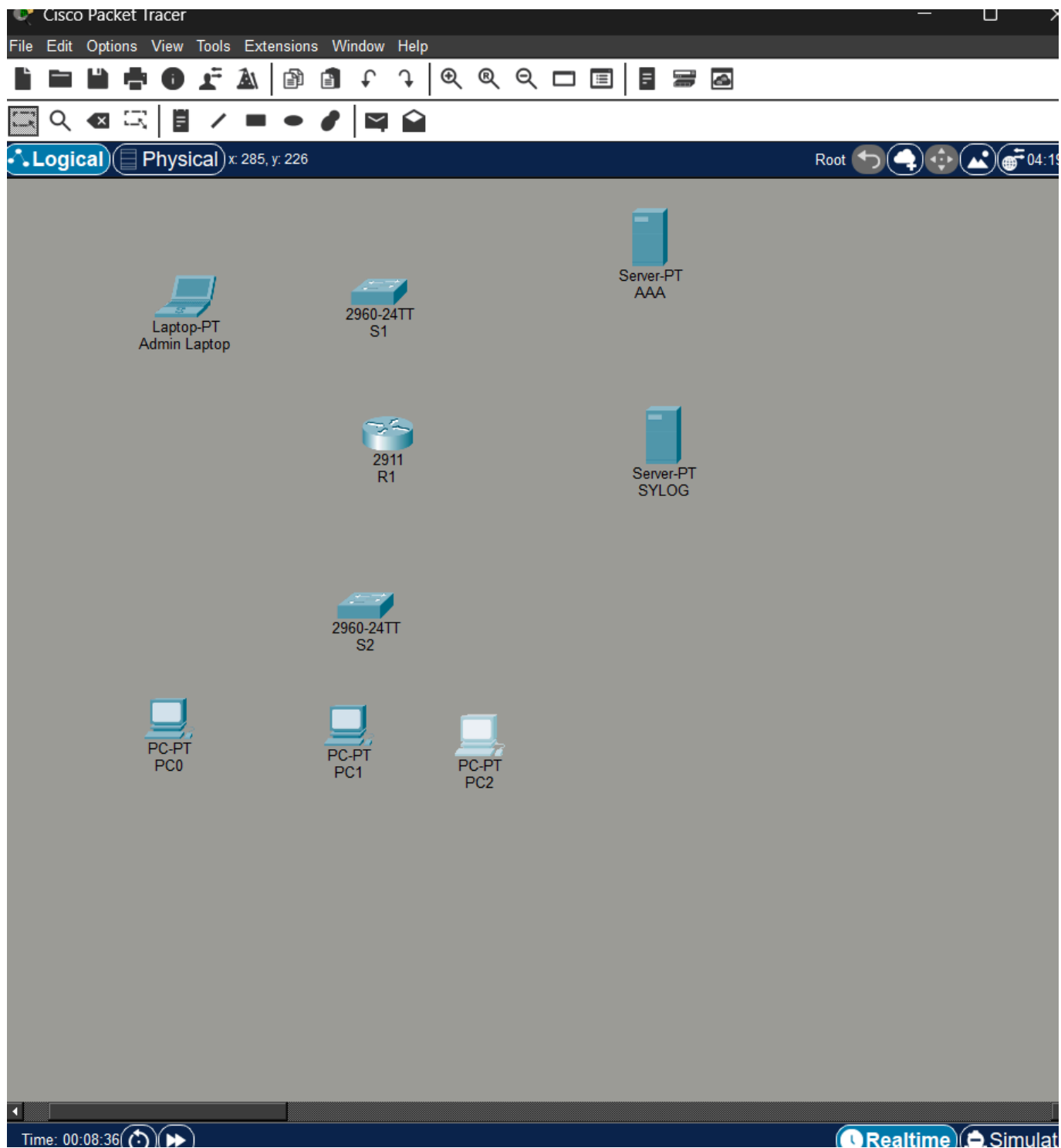
- Cable: **Serial DCE**
- Click **R2** → select **Serial0/0/1** (DCE end connects here)
- Click **R3** → select **Serial0/0/0**

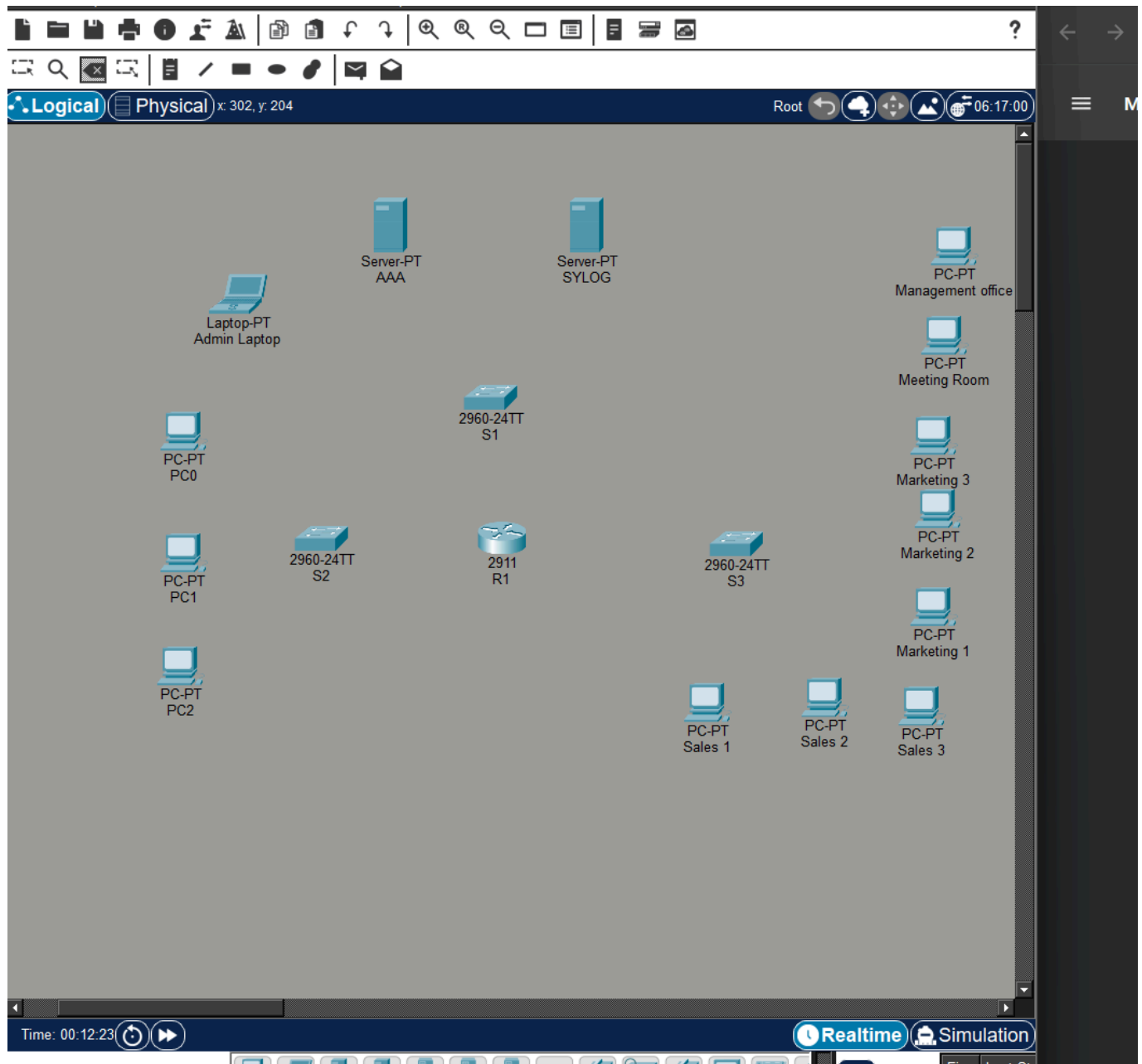
EXTERNAL / INTERNET connections:

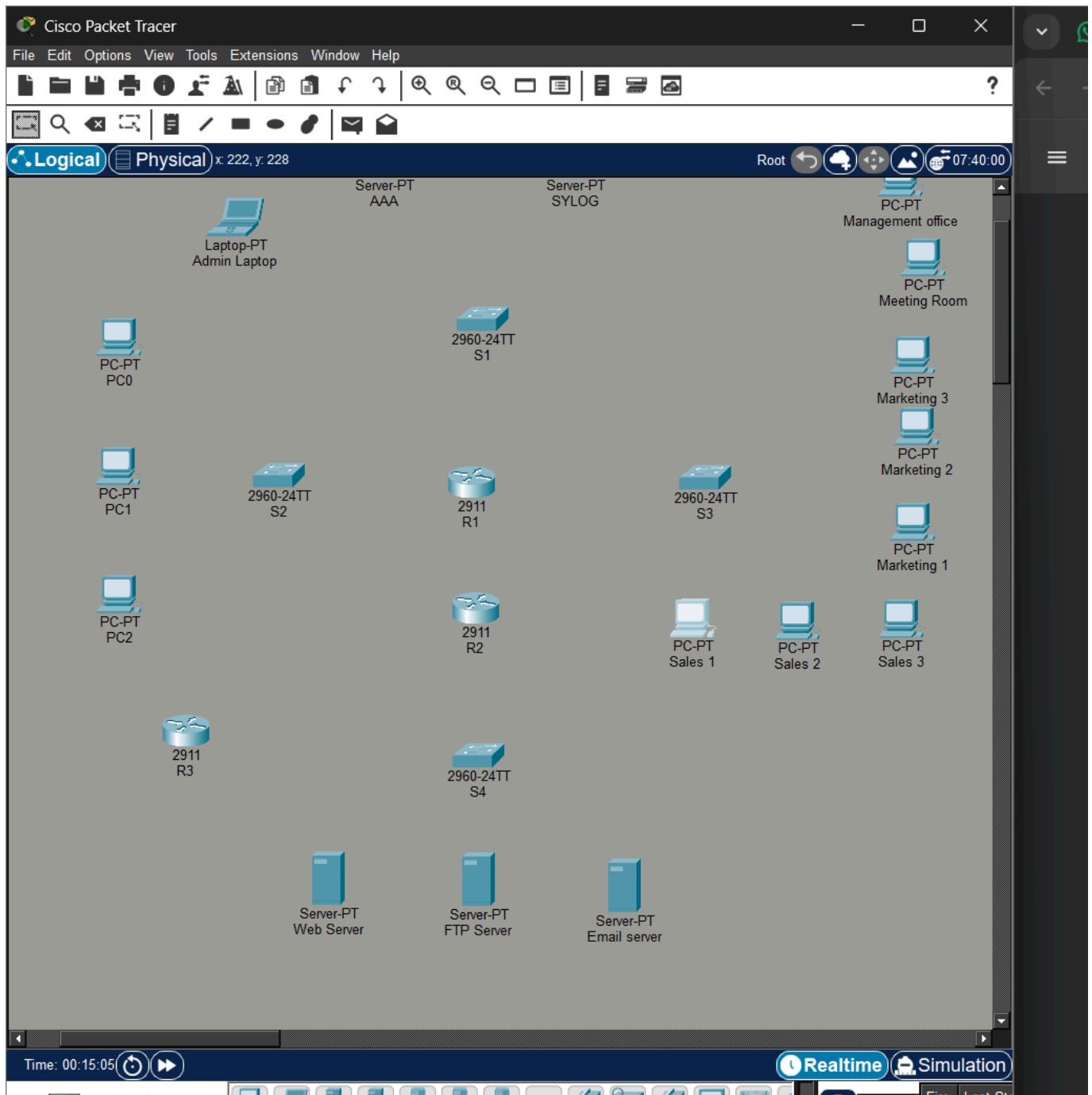
Connection 25 – R3 to Internet-Server

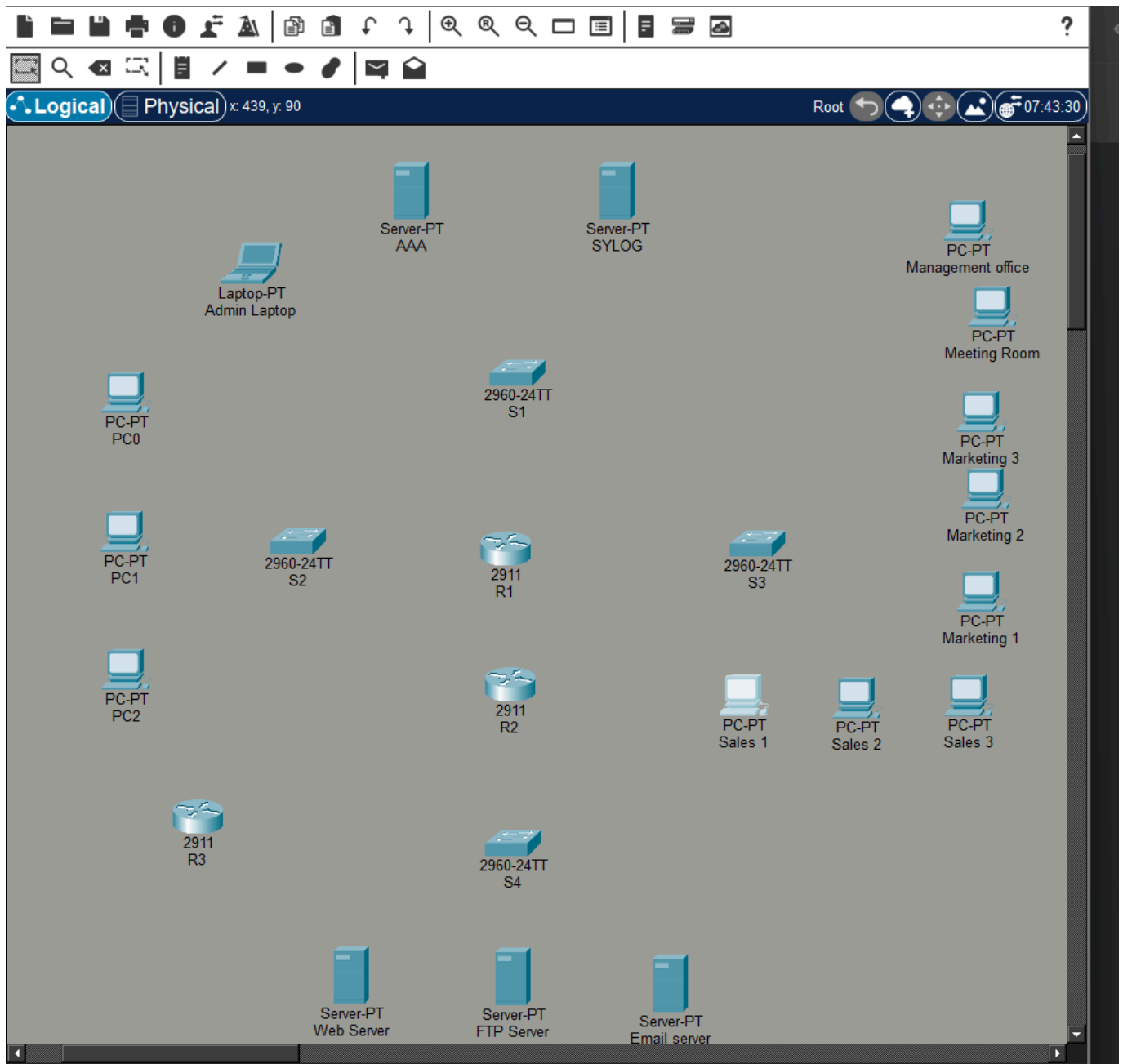
- Cable: **Copper Straight-Through**
- Click **R3** → select **GigabitEthernet0/0**
- Click **Internet-Server** → select **FastEthernet0**

At this point, many cable dots will be red (link down). This is expected — routers need IP configuration before links turn green.









5. Part B – Configuring Router R1 (Server Room)

To access R1's CLI: **Double-click R1** → **CLI tab** → Press **Enter**.

Type **no** if prompted: "Would you like to enter the initial configuration dialog?"

Step B1 – Basic Hostname and Password Setup

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# service password-encryption
R1(config)# enable secret cisco123Pa55
```

`enable secret` stores the password as an MD5 hash — it is already hashed/encrypted. `service password-encryption` encrypts all other plain-text passwords (Type 7).

Step B2 – Create Local User Accounts

```
R1(config)# username Admin1 privilege 15 secret admin1Pa55
R1(config)# username SSHAdmin1 privilege 15 secret SSHadmin1Pa55
```

Using `secret` (not `password`) ensures each user password is hashed.

Step B3 – Configure Console Line (Local Access via Console Cable)

```
R1(config)# line console 0
R1(config-line)# login local
R1(config-line)# exec-timeout 5 0
R1(config-line)# exit
```

`login local` uses the local username/password database (Admin1 / admin1Pa55 for console access).
`exec-timeout 5 0` = auto-logout after 5 minutes of inactivity.

Step B4 – Configure VTY Lines (Remote SSH Access)

```
R1(config)# line vty 0 4
R1(config-line)# login local
R1(config-line)# transport input ssh
R1(config-line)# exec-timeout 5 0
R1(config-line)# exit
```

`transport input ssh` restricts VTY to SSH only — Telnet is blocked. `login local` uses SSHAdmin1 / SSHadmin1Pa55 for SSH remote access.

Step B5 – Enable SSH

```
R1(config)# ip domain-name company.local
R1(config)# crypto key generate rsa
```

When prompted "**How many bits in the modulus [512]:**" type `1024` and press Enter.

```
R1(config)# ip ssh version 2
```

Step B6 – Configure Interfaces

GigabitEthernet0/0 – Server Room (192.168.1.0/24)

```
R1(config)# interface GigabitEthernet0/0
R1(config-if)# description Server_Room_LAN
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
```

GigabitEthernet0/1 – Technical Support (192.168.2.0/24)

```
R1(config)# interface GigabitEthernet0/1
R1(config-if)# description Tech_Support_LAN
R1(config-if)# ip address 192.168.2.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
```

GigabitEthernet0/2 – Employees (192.168.3.0/24)

```
R1(config)# interface GigabitEthernet0/2
R1(config-if)# description Employees_LAN
R1(config-if)# ip address 192.168.3.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
```

Serial0/0/0 – WAN to R2 (172.16.1.0/24) — R1 is DCE

```
R1(config)# interface Serial0/0/0
R1(config-if)# description WAN_Link_to_R2
R1(config-if)# ip address 172.16.1.1 255.255.255.0
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
R1(config-if)# exit
```

`clock rate 64000` is required on the **DCE** end only. R1 is DCE on this link.

Press RETURN to get started!

Press RETURN to get started!

Press RETURN to get started!

```
Router>
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
R1(config)#enable secret Strong123
R1(config)#service password-encryption
R1(config)#
```

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```
Router>
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
R1(config)#enable secret Strong123
R1(config)#service password-encryption
R1(config)#interface g0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down

R1(config-if)#
```

[Copy](#)[Top](#)

```
R1(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down

R1(config-if)#interface g0/1
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down

R1(config-if)#
```

[Copy](#)[Top](#)

```
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down

R1(config-if)#interface g0/2
R1(config-if)#ip address 192.168.3.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to down

R1(config-if)#
```

Copy

Top

```
Device      Label in PT  Interface  IP Address  Status  Protocol
-----
R1          S1          S1          192.168.1.1  up      up
R1          S2          S2          192.168.2.1  up      up
R1          S3          S3          192.168.3.1  up      up
R1          S4          S4          unassigned  down    down
```

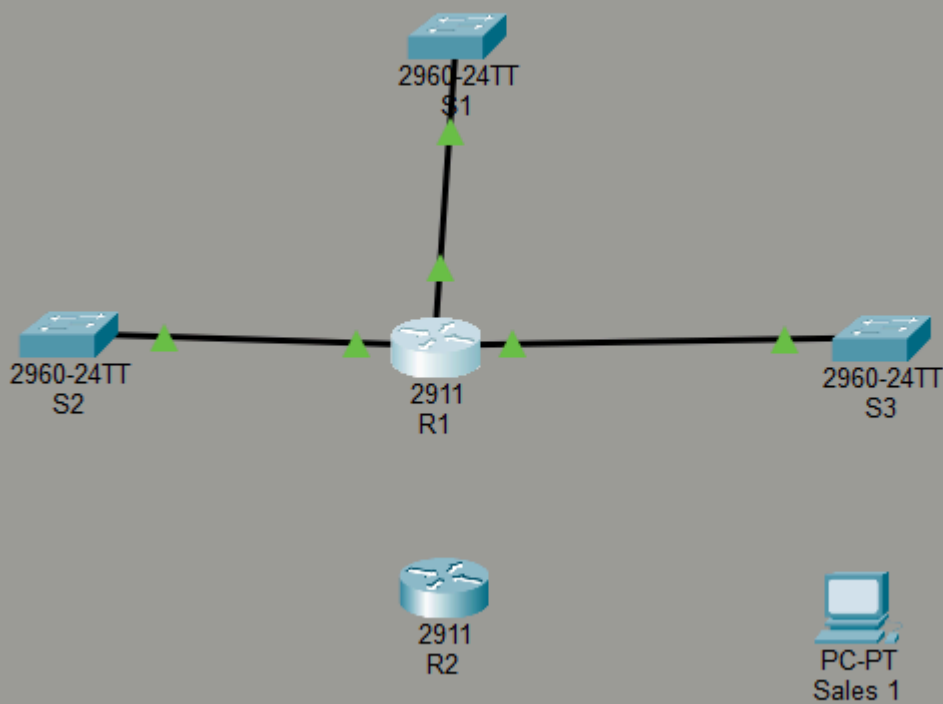
R1#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.1.1	YES	manual	up	up
GigabitEthernet0/1	192.168.2.1	YES	manual	up	up
GigabitEthernet0/2	192.168.3.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

R1#

Copy

Laptop-PT
Admin Laptop



Step B7 – Configure Brute-Force Login Protection

```
R1(config)# login block-for 300 attempts 3 within 60
```

This blocks any login for **300 seconds (5 minutes)** if there are **3 failed attempts within 60 seconds (1 minute)** — exactly as required.

Step B8 – Configure MOTD Banner

```
R1(config)# banner motd # WARNING: Unauthorized access is strictly prohibited! All  
activity is monitored and logged. #
```

6. Part C – Configuring Router R2 (Edge/DMZ)

Double-click R2 → **CLI tab** → Press **Enter**.

Step C1 – Basic Hostname and Password Setup

```
Router> enable  
Router# configure terminal  
Router(config)# hostname R2  
R2(config)# service password-encryption  
R2(config)# enable secret cisco123Pa55
```

Step C2 – Create Local User Accounts

```
R2(config)# username Admin1 privilege 15 secret admin1Pa55  
R2(config)# username SSHAdmin1 privilege 15 secret SSHAdmin1Pa55
```

Step C3 – Configure Console Line

```
R2(config)# line console 0  
R2(config-line)# login local  
R2(config-line)# exec-timeout 5 0  
R2(config-line)# exit
```

Step C4 – Configure VTY Lines (SSH)

```
R2(config)# line vty 0 4
R2(config-line)# login local
R2(config-line)# transport input ssh
R2(config-line)# exec-timeout 5 0
R2(config-line)# exit
```

Step C5 – Enable SSH

```
R2(config)# ip domain-name company.local
R2(config)# crypto key generate rsa
```

Type **1024** when prompted for modulus size.

```
R2(config)# ip ssh version 2
```

Step C6 – Configure Interfaces

Serial0/0/0 – WAN from R1 (172.16.1.0/24) — R2 is DTE (NO clock rate needed)

```
R2(config)# interface Serial0/0/0
R2(config-if)# description WAN_Link_from_R1
R2(config-if)# ip address 172.16.1.2 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
```

GigabitEthernet0/0 – DMZ Network (172.16.3.0/24)

```
R2(config)# interface GigabitEthernet0/0
R2(config-if)# description DMZ_Network
R2(config-if)# ip address 172.16.3.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
```

Serial0/0/1 – WAN to R3 (172.16.2.0/24) — R2 is DCE

```
R2(config)# interface Serial0/0/1
R2(config-if)# description WAN_Link_to_R3
R2(config-if)# ip address 172.16.2.1 255.255.255.0
R2(config-if)# clock rate 64000
```

```
R2(config-if)# no shutdown
R2(config-if)# exit
```

R2 is DCE on the R2→R3 link, so **clock rate 64000** is needed here.

Step C7 – Configure Brute-Force Login Protection

```
R2(config)# login block-for 300 attempts 3 within 60
```

Step C8 – Configure MOTD Banner

```
R2(config)# banner motd # WARNING: Unauthorized access is strictly prohibited! All
activity is monitored and logged. #
```

7. Part D – Configuring Router R3 (ISP/External)

Double-click R3 → **CLI tab** → Press **Enter**.

Step D1 – Basic Hostname and Password Setup

```
Router> enable
Router# configure terminal
Router(config)# hostname R3
R3(config)# service password-encryption
R3(config)# enable secret cisco123Pa55
```

Step D2 – Create Local User Accounts

```
R3(config)# username Admin1 privilege 15 secret admin1Pa55
R3(config)# username SSHAdmin1 privilege 15 secret SSHadmin1Pa55
```

Step D3 – Configure Console Line

```
R3(config)# line console 0
R3(config-line)# login local
R3(config-line)# exec-timeout 5 0
R3(config-line)# exit
```

Step D4 – Configure VTY Lines (SSH)

```
R3(config)# line vty 0 4
R3(config-line)# login local
R3(config-line)# transport input ssh
R3(config-line)# exec-timeout 5 0
R3(config-line)# exit
```

Step D5 – Enable SSH

```
R3(config)# ip domain-name company.local
R3(config)# crypto key generate rsa
```

Type **1024** when prompted.

```
R3(config)# ip ssh version 2
```

Step D6 – Configure Interfaces

Serial0/0/0 – WAN from R2 (172.16.2.0/24) — R3 is DTE (NO clock rate needed)

```
R3(config)# interface Serial0/0/0
R3(config-if)# description WAN_Link_from_R2
R3(config-if)# ip address 172.16.2.2 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# exit
```

GigabitEthernet0/0 – Simulated Internet (209.165.200.0/24)

```
R3(config)# interface GigabitEthernet0/0
R3(config-if)# description Simulated_Internet
R3(config-if)# ip address 209.165.200.1 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# exit
```

Step D7 – Configure Brute-Force Login Protection

```
R3(config)# login block-for 300 attempts 3 within 60
```

Step D8 – Configure MOTD Banner

```
R3(config)# banner motd # WARNING: Unauthorized access is strictly prohibited! All activity is monitored and logged. #
```

8. Part E – Configuring End-Device IP Addresses

For each end device below, **double-click the device** → **Desktop tab** → **IP Configuration**. Set each field manually (Static). Repeat for every device.

For **Server-PT** devices: Double-click → **Config tab** → **FastEthernet0** → enter IP/Mask/Gateway. OR: Desktop tab → IP Configuration (same result).

Server Room Devices

AAA-Server:

- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- DNS Server: 192.168.1.2 (optional, can leave blank)

SYSLOG-Server:

- IP Address: 192.168.1.3
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

Admin-Laptop:

- IP Address: 192.168.1.4
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1
-

Technical Support PCs

IT-PC1:

- IP Address: 192.168.2.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

IT-PC2:

- IP Address: 192.168.2.3
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

IT-PC3:

- IP Address: 192.168.2.4
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

```

R3
Physical Config CLI Attributes

DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!

Router>conf t
      ^
% Invalid input detected at '^' marker.

Router>
Router>interface s0/0/0
      ^
% Invalid input detected at '^' marker.

Router>ip address 172.16.2.2 255.255.255.0
      ^
% Invalid input detected at '^' marker.

Router>configure
Translating "configure"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#conf t
%Invalid hex value
Router(config)#
Router(config)#interface s0/0/0
Router(config-if)#ip address 172.16.2.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

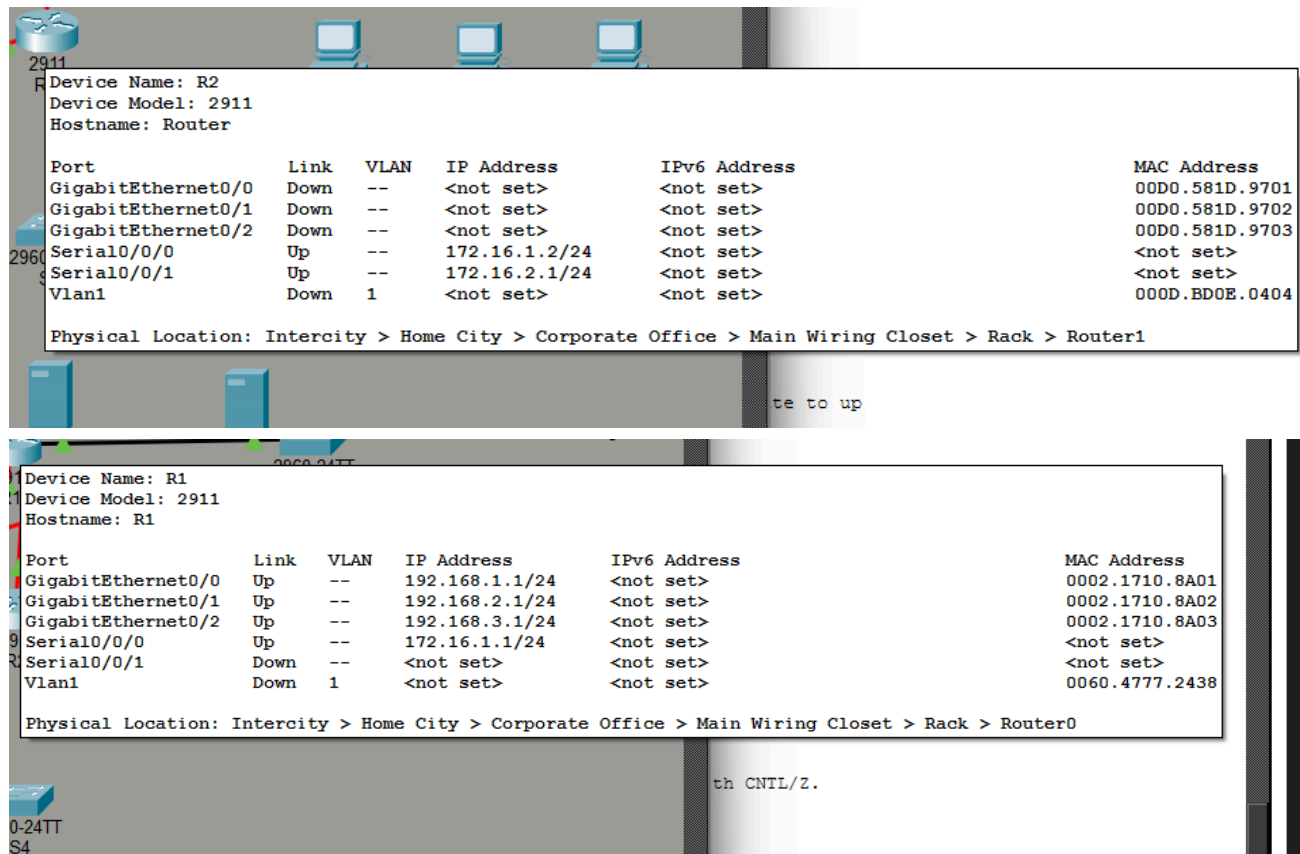
```

2911 R3

Device Name: R3
Device Model: 2911
Hostname: Router

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
GigabitEthernet0/0	Down	--	<not set>	<not set>	0030.F24B.E101
GigabitEthernet0/1	Down	--	<not set>	<not set>	0030.F24B.E102
GigabitEthernet0/2	Down	--	<not set>	<not set>	0030.F24B.E103
Serial0/0/0	Up	--	172.16.2.2/24	<not set>	<not set>
Serial0/0/1	Down	--	<not set>	<not set>	<not set>
Vlan1	Down	1	<not set>	<not set>	00D0.97D6.0DD0

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router2



Employees' Office PCs

Sales-PC1:

- IP Address: 192.168.3.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.3.1

Sales-PC2:

- IP Address: 192.168.3.3
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.3.1

Sales-PC3:

- IP Address: 192.168.3.4
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.3.1

AAA

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

Subnet Mask

Default Gateway0.0.0.0

DNS Server

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local AddressFE80::2E0:B0FF:FEE8:CB35

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

AuthenticationMD5

Username

Password

AAA

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration

IP Configuration

DHCP

Static

IPv4 Address

192.168.1.3

Subnet Mask

Default Gateway

0.0.0.0

DNS Server

192.168.1.1

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::2E0:B0FF:FEE8:CB35

Default Gateway

DNS Server

802.1X

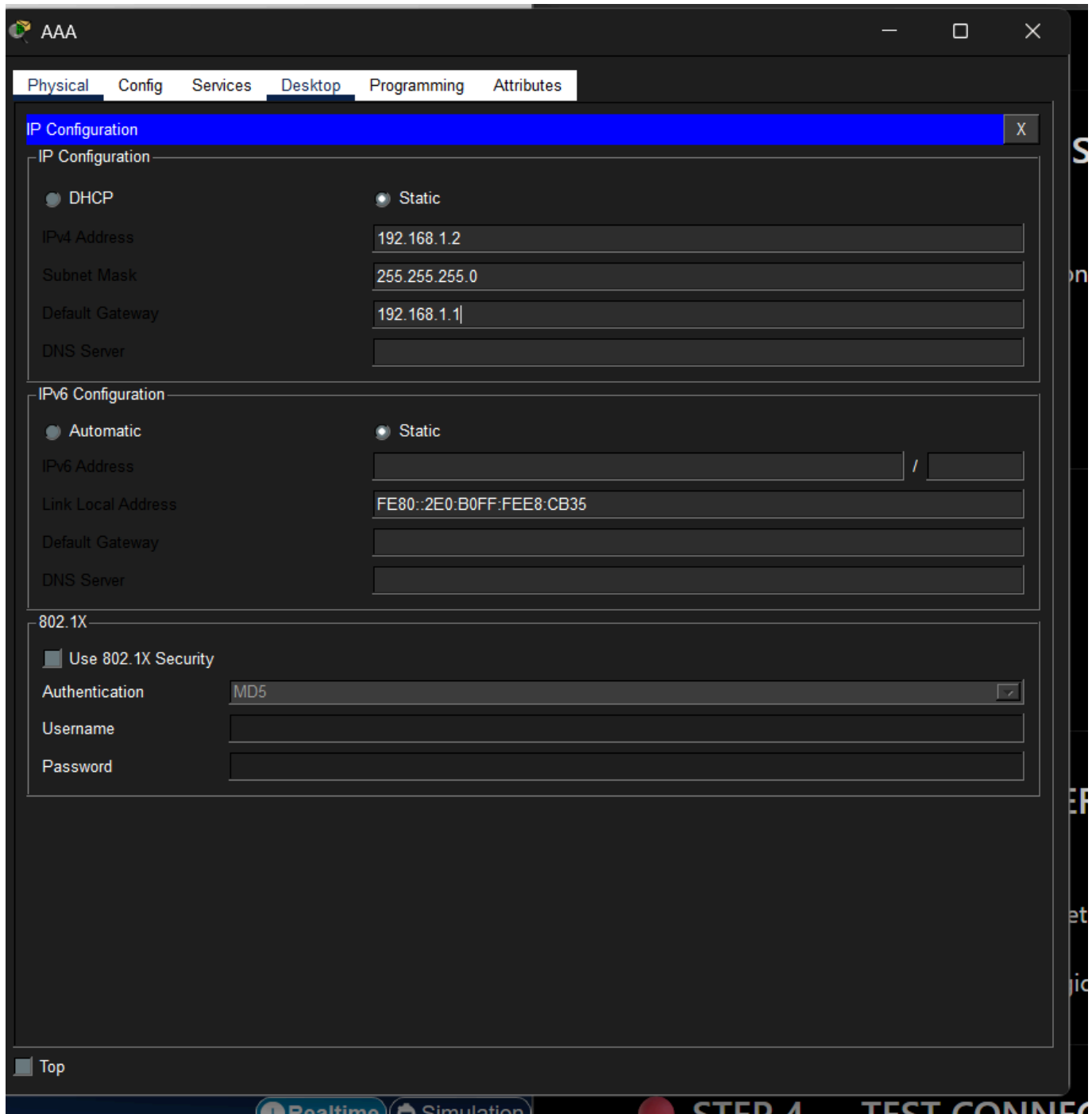
Use 802.1X Security

Authentication

MD5

Username

Password

**Mktg-PC1:**

- IP Address: 192.168.3.5
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.3.1

Mktg-PC2:

- IP Address: 192.168.3.6
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.3.1

Mktg-PC3:

- IP Address: 192.168.3.7

- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.3.1

SYLOG

Physical

Config

Services

Desktop

Programming

Attributes

IP Configuration

X

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.1.3

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::20A:41FF:FEDD:6681

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

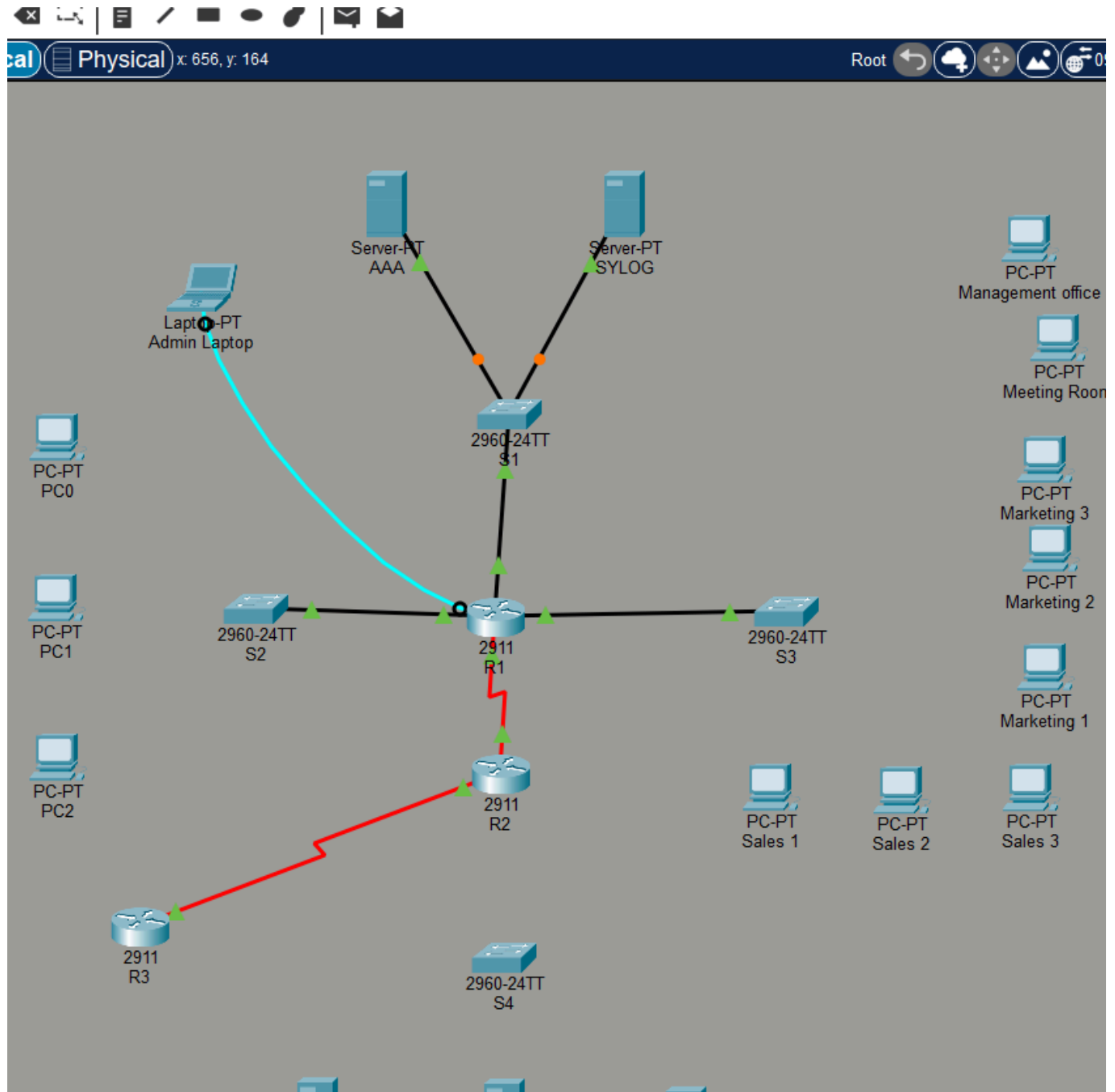
SERV

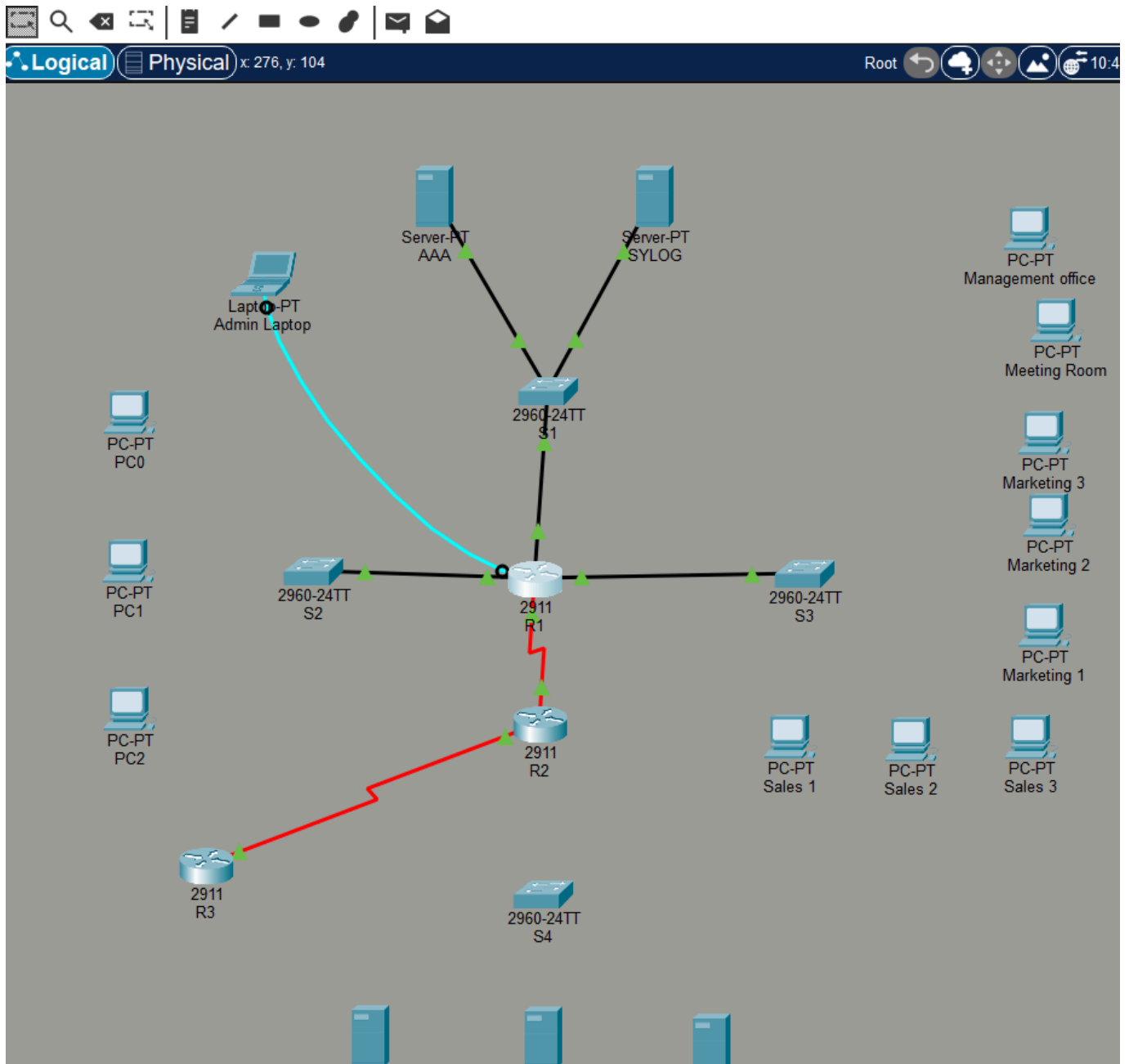
on:

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ically





Meeting-PC:

- IP Address: 192.168.3.8
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.3.1

Mgmt-PC:

- IP Address: 192.168.3.9
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.3.1

```
R1>
R1>ping 192.168.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

R1>
R1>ping 192.168.1.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.3, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

R1>
```

Cop

Press RETURN to get started.

```
R1>
R1>ping 192.168.1.2 (AAA)
^
% Invalid input detected at '^' marker.

R1>ping 192.168.1.3 (SYSLOG)
R1>
R1>
R1>enable
Password:
Password:
Password:
% Bad secrets

R1>
R1>ping 192.168.1.2

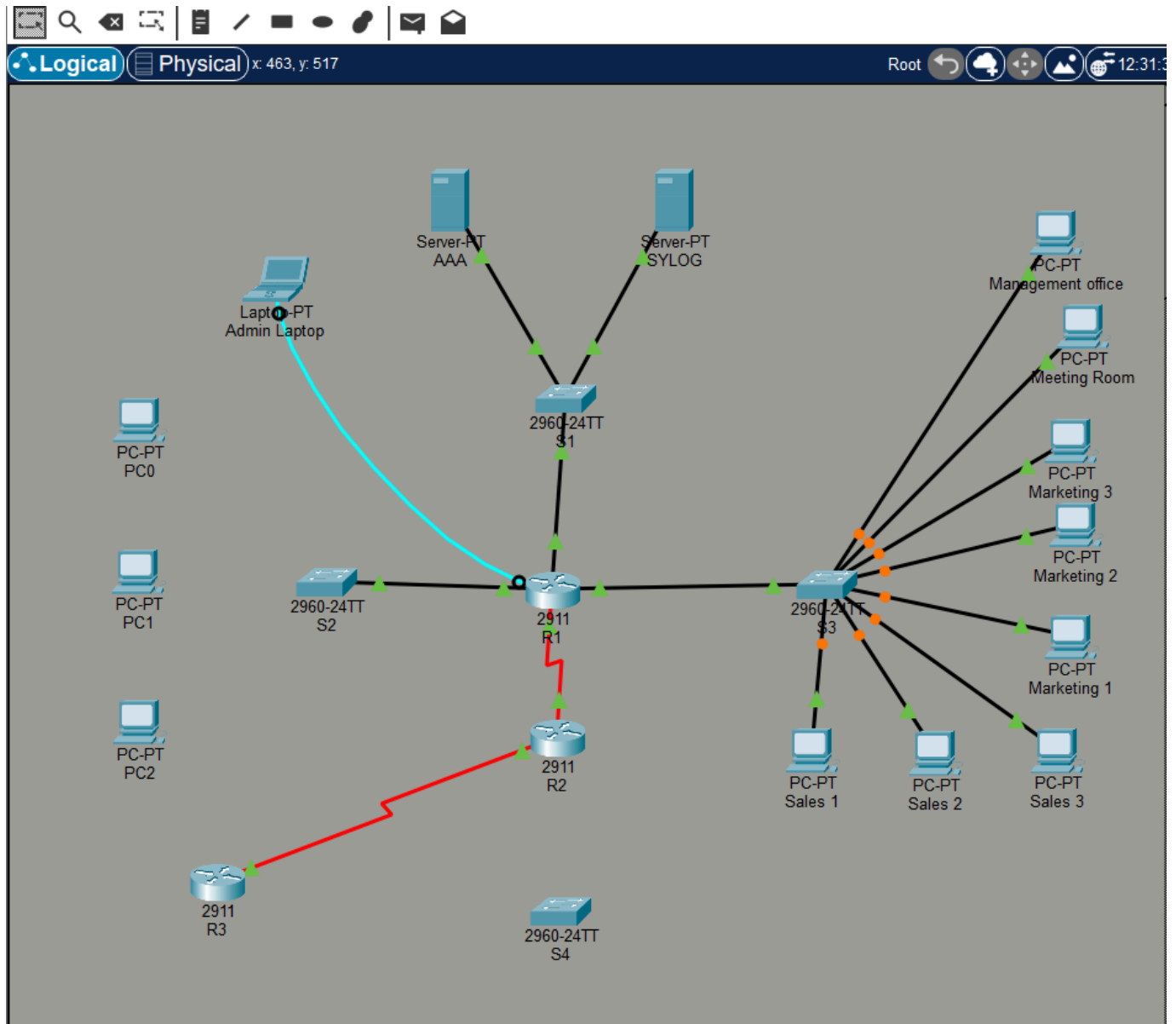
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

R1>
R1>ping 192.168.1.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.3, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

R1>
```

Copy Paste



DMZ Servers

Web-Server:

- IP Address: 172.16.3.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 172.16.3.1
- Also: Double-click Web-Server → **Services tab** → Click **HTTP** → Ensure it is **ON**.

FTP-Server:

- IP Address: 172.16.3.3
- Subnet Mask: 255.255.255.0
- Default Gateway: 172.16.3.1
- Also: Double-click FTP-Server → **Services tab** → Click **FTP** → Ensure it is **ON**.

Email-Server:

- IP Address: 172.16.3.4
- Subnet Mask: 255.255.255.0

- Default Gateway: **172.16.3.1**
 - Also: Double-click Email-Server → **Services tab** → Click **EMAIL** → Ensure it is **ON**.
-

Internet Server (Simulated External Network)

Internet-Server:

- IP Address: **209.165.200.2**
- Subnet Mask: **255.255.255.0**
- Default Gateway: **209.165.200.1**

Sales 3

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.3.4

Subnet Mask

255.255.255.0

Default Gateway

192.168.3.1

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::20B:BEFF:FE1E:C0BB

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

ation

X

FastEthernet0

uration

Static

192.168.3.3

255.255.255.0

192.168.3.1

0.0.0.0

guration

Static

/

FE80::2D0:BCFF:FEB8:E6AE

802.1X Security

MD5

Sales 2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.3.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.3.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:BCFF:FEB8:E6AE

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Top

Marketing 3

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

X

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address192.168.3.7

Subnet Mask255.255.255.0

Default Gateway192.168.3.1

DNS Server0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address /

Link Local AddressFE80::20C:CFFF:FEE1:54AE

Default Gateway

DNS Server

802.1X

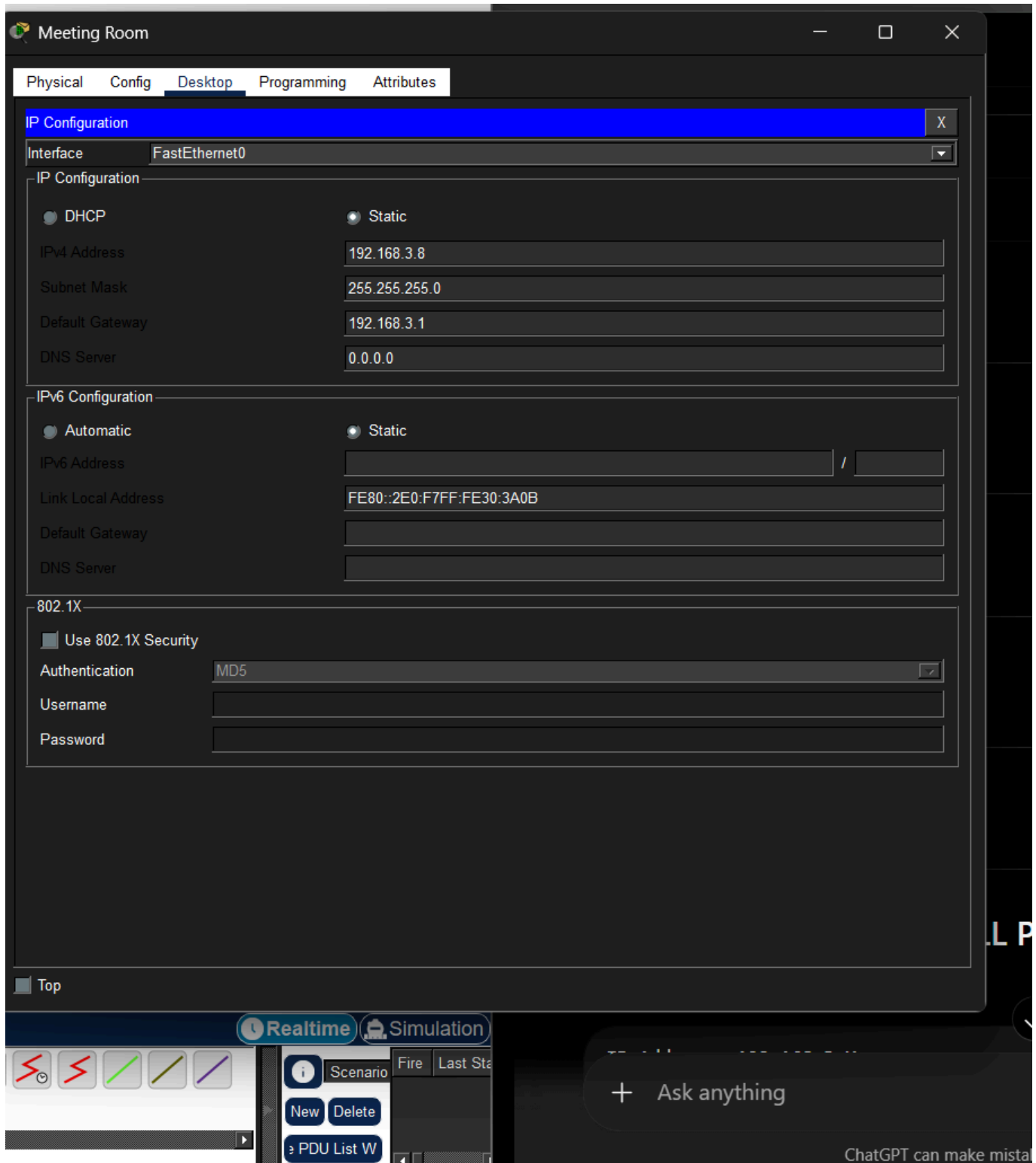
☐ Use 802.1X Security

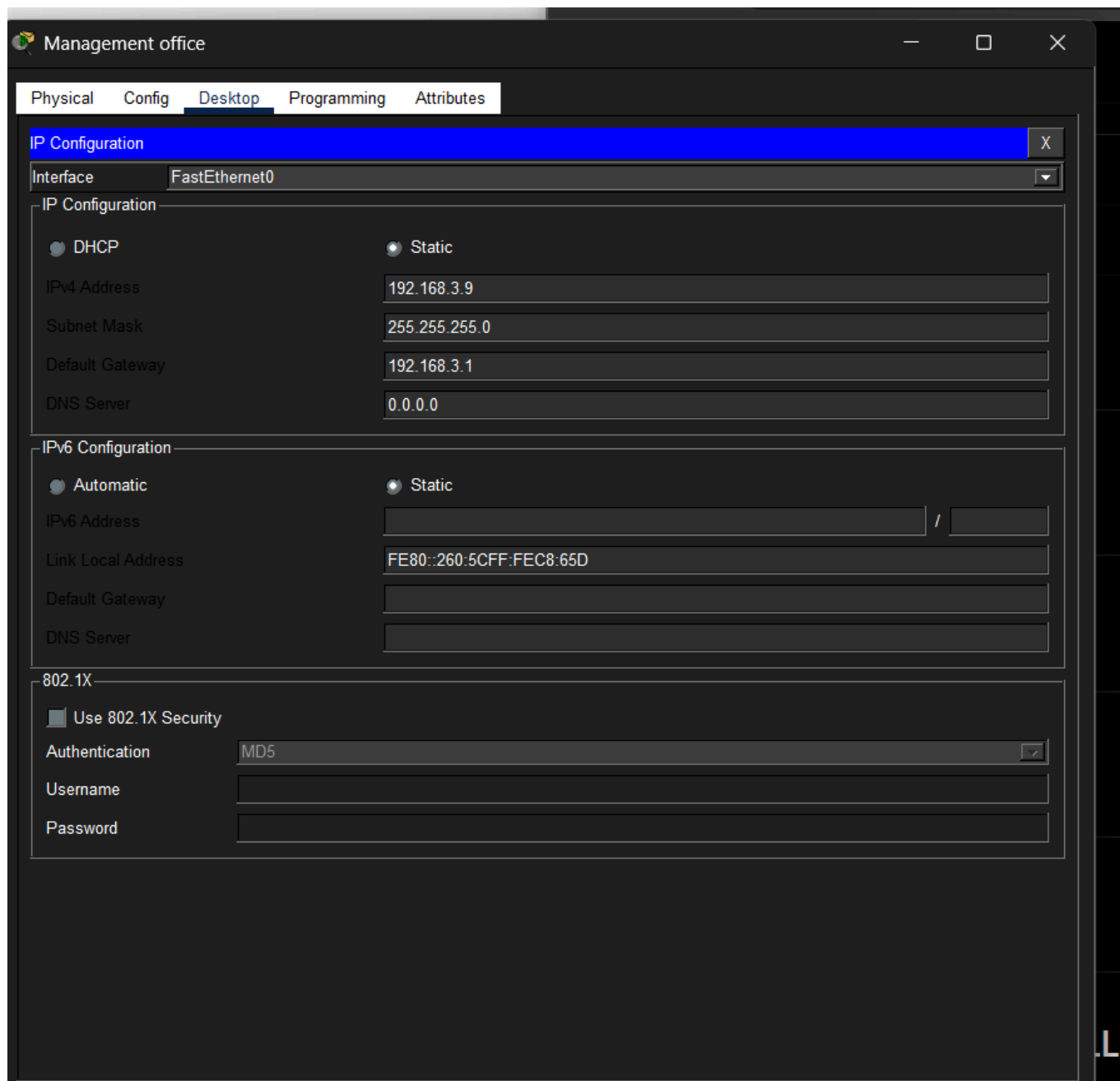
AuthenticationMD5

Username

Password

Top





```
.!!!!  
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms  
  
R1>  
R1>ping 192.168.1.3  
  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 192.168.1.3, timeout is 2 seconds:  
.!!!!  
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms  
  
R1>ping 192.168.3.1  
  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:  
!!!!!  
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/9/22 ms  
  
R1>
```

[Copy](#)

9. Part F – Configuring the AAA (TACACS+) Server

The AAA server runs the TACACS+ service. All routers will authenticate login through this server.

Double-click AAA-Server → Services tab → AAA (in the left service list).

Step F1 – Enable AAA Service

- Set the service toggle to **ON**.

Step F2 – Add Network Clients (Routers)

Under "**Network Configuration**", click "**Add**" for each router:

Add R1 as client:

- Client Name: **R1**
- Client IP: **192.168.1.1**
- Secret: **TacacsKey123**
- ServerType: **TACACS+**
- Click **Add**

Add R2 as client:

- Client Name: **R2**
- Client IP: **172.16.1.2**
- Secret: **TacacsKey123**
- ServerType: **TACACS+**
- Click **Add**

Add R3 as client:

- Client Name: **R3**
- Client IP: **172.16.2.2**
- Secret: **TacacsKey123**
- ServerType: **TACACS+**
- Click **Add**

Step F3 – Add Users to AAA Server

Under "**User Setup**", click "**Add**" for each user:

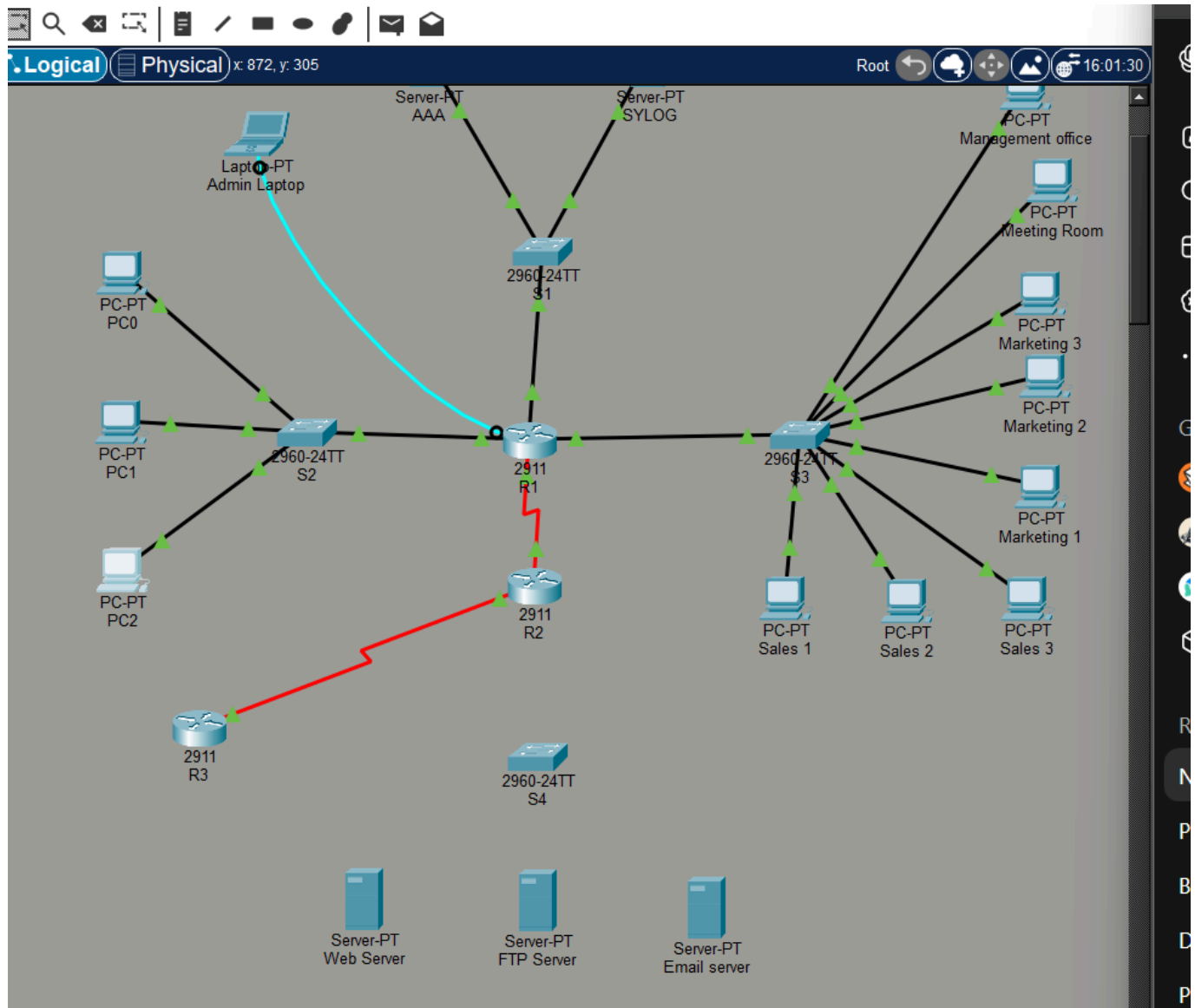
Add Admin1:

- Username: **Admin1**
- Password: **admin1Pa55**
- Click **Add**

Add SSHAdmin1:

- Username: **SSHAdmin1**
- Password: **SSHadmin1Pa55**

- Click **Add**



PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.2.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:FFFF:FE4A:3635

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Top

PC1

Physical

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Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

192.168.2.3

Subnet Mask

255.255.255.0

Default Gateway

192.168.2.1

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::250:FFF:FE87:9EDB

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.2.4

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20A:41FF:FEEE:1543

Default Gateway

DNS Server

802.1X

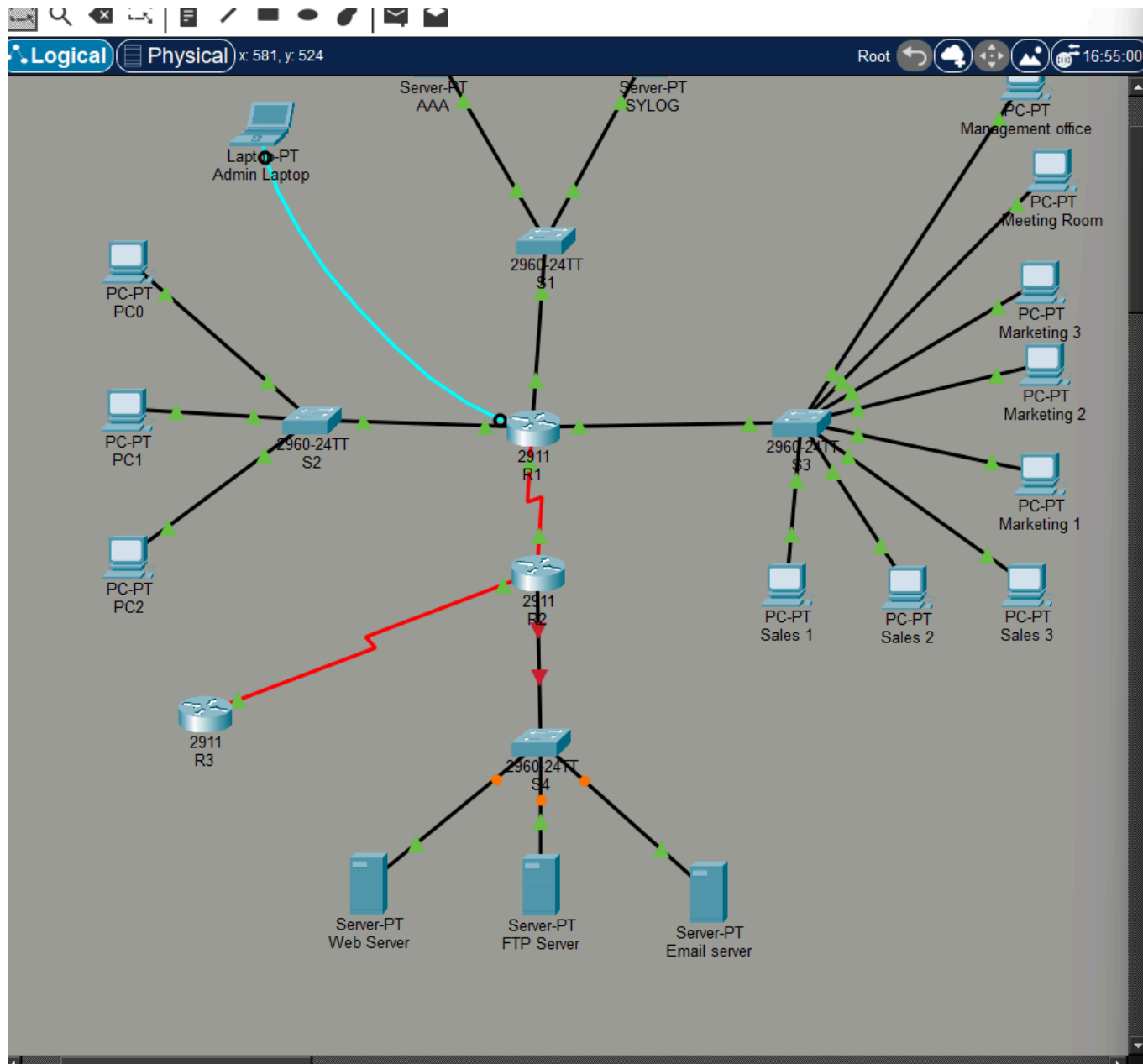
☐ Use 802.1X Security

Authentication MD5

Username

Password

Top



10. Part G – Configuring the SYSLOG Server

Double-click SYSLOG-Server → **Services tab** → **SYSLOG** (in the left service list).

- Set the service toggle to **ON**.

The SYSLOG server will now collect all log messages sent to IP **192.168.1.3** from the routers (configured in Part J).

Physical

Config

Services

Desktop

Programming

Attributes

IP Configuration

DHCP

Static

IPv4 Address

172.16.3.2

Subnet Mask

Default Gateway

172.16.3.1

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address /

Link Local Address

FE80::209:7CFF:FE89:6237

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Web Server

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration

DHCP

Static

IPv4 Address

172.16.3.2

Subnet Mask

255.255.0.0

Default Gateway

172.16.3.1

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local Address

FE80::209:7CFF:FE89:6237

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

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Realtime

Simulation

CSV ERP Architecture

Gatewa

FTP Server

PhysicalConfigServicesDesktopProgrammingAttributes

IP ConfigurationX

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

172.16.3.3

Subnet Mask

Default Gateway

172.16.3.1

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::209:7CFF:FECD:4516

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

Top

 R2

Physical Config CLI Attributes

```
Router>conf t
      ^
% Invalid input detected at '^' marker.

Router>
Router>interface g0/0
      ^
% Invalid input detected at '^' marker.

Router>ip address 172.16.3.1 255.255.255.0
      ^
% Invalid input detected at '^' marker.

Router>no shutdown
Translating "no"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router>
Router>
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#conf t
%Invalid hex value
Router(config)#
Router(config)#interface g0/0
Router(config-if)#ip address 172.16.3.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

Copy Paste

 Top



OSPF on R1

```
R1(config)# router ospf 1
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
R1(config-router)# network 192.168.2.0 0.0.0.255 area 0
R1(config-router)# network 192.168.3.0 0.0.0.255 area 0
R1(config-router)# network 172.16.1.0 0.0.0.255 area 0
R1(config-router)# exit
```

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OSPF on R2

In R2's CLI:

```
R2(config)# router ospf 1
R2(config-router)# network 172.16.1.0 0.0.0.255 area 0
R2(config-router)# network 172.16.2.0 0.0.0.255 area 0
R2(config-router)# network 172.16.3.0 0.0.0.255 area 0
R2(config-router)# exit
```

OSPF on R3

In R3's CLI:

```
R3(config)# router ospf 1
R3(config-router)# network 172.16.2.0 0.0.0.255 area 0
R3(config-router)# network 209.165.200.0 0.0.0.255 area 0
R3(config-router)# exit
```

After a few seconds, you should see OSPF neighbor messages appear in the CLI (e.g., **%OSPF-5-ADJCHG: Process 1, Nbr ... State FULL**). This confirms OSPF neighbors are forming correctly.

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#ping 172.16.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/4/10 ms
Router#
```

[Copy](#)[Top](#)

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/4/10 ms

Router#ping 172.16.3.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.3.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/2/8 ms
Router#
```

[Co](#)

```
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 172.16.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/4/10 ms

Router#ping 172.16.3.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.3.2, timeout is 2 seconds:
!!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/2/8 ms

Router#ping 172.16.3.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.3.3, timeout is 2 seconds:
!!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 172.16.3.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.3.4, timeout is 2 seconds:
!!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/3/13 ms

Router#
```

Copy

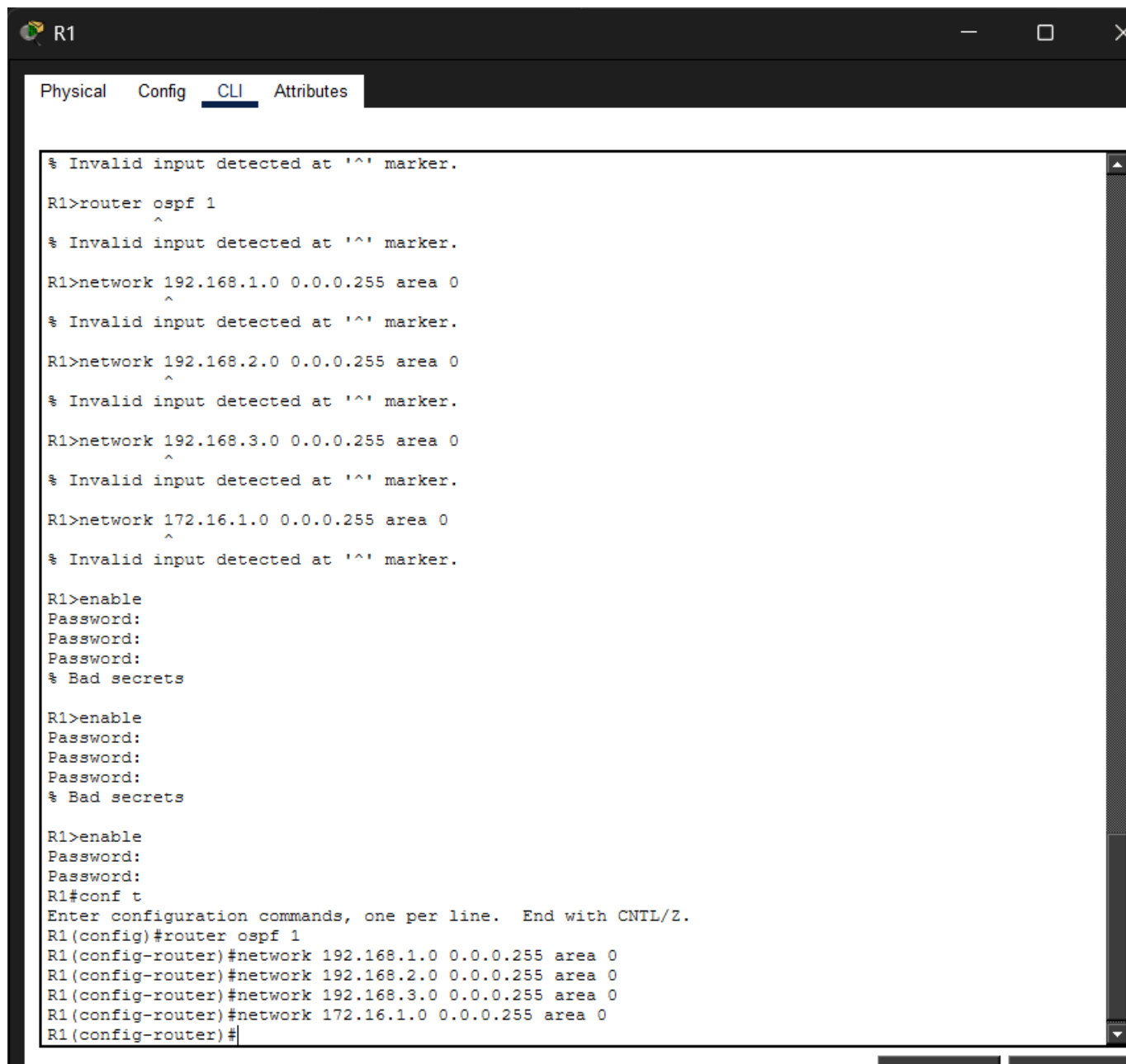
Paste

The screenshot shows the Cisco Packet Tracer application window. At the top, there is a 'Top' button. The main area contains a terminal window with the following text:

```
Router>
Router>enable
Router#configure
Configuring from terminal, memory, or network [terminal]? terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 209.165.200.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down
```

Below the terminal window, there are 'Copy' and 'Paste' buttons. At the bottom of the application, there is a toolbar with 'Realtime' and 'Simulation' tabs, and a status bar showing 'CSV ERP Architecture' and 'STEP'.



The image shows a Cisco Packet Tracer window for router R1. The 'CLI' tab is selected, displaying a command-line interface. The user has entered several commands to configure OSPF, but each line is followed by an error message: '% Invalid input detected at '^' marker.' The commands entered are:

```
R1>router ospf 1
R1>network 192.168.1.0 0.0.0.255 area 0
R1>network 192.168.2.0 0.0.0.255 area 0
R1>network 192.168.3.0 0.0.0.255 area 0
R1>network 172.16.1.0 0.0.0.255 area 0
```

After these commands, the user enters 'enable' three times, each followed by 'Password:' and then '% Bad secrets'.

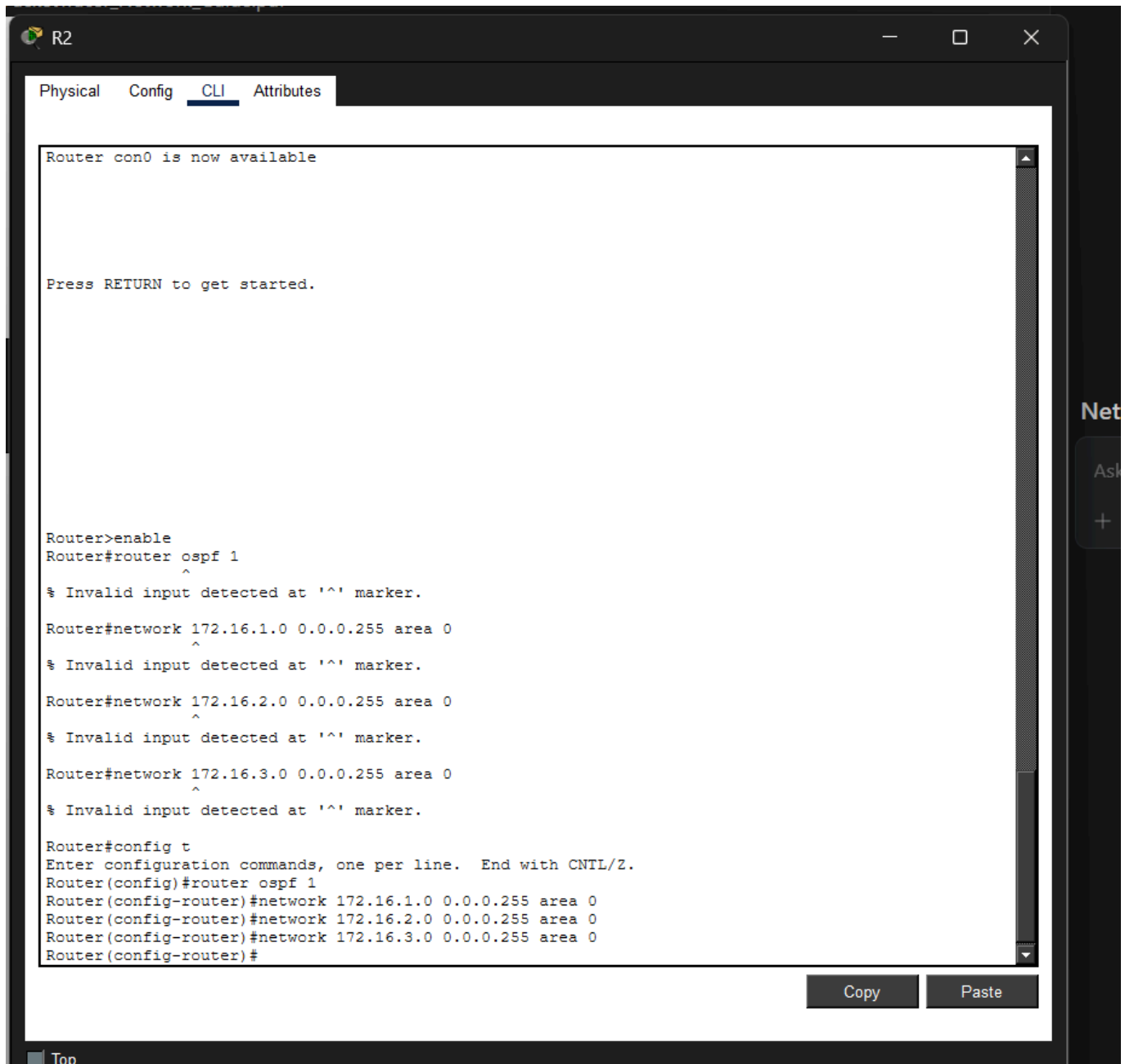
```
R1>enable
Password:
Password:
Password:
% Bad secrets

R1>enable
Password:
Password:
Password:
% Bad secrets

R1>enable
Password:
Password:
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
```

Finally, the user enters configuration mode and enters the OSPF configuration commands correctly:

```
R1(config)#router ospf 1
R1(config-router)#network 192.168.1.0 0.0.0.255 area 0
R1(config-router)#network 192.168.2.0 0.0.0.255 area 0
R1(config-router)#network 192.168.3.0 0.0.0.255 area 0
R1(config-router)#network 172.16.1.0 0.0.0.255 area 0
R1(config-router)#
```



 R3

Physical Config CLI Attributes

```
Configuring from terminal, memory, or network [terminal]? terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 209.165.200.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down

Router con0 is now available

Press RETURN to get started.

Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 172.16.2.0 0.0.0.255 area 0
Router(config-router)#network 209.165.200.0 0.0.0.255 area 0
Router(config-router)#
```

Copy Paste

Top

area 0

on Serial0/0/0 from LOADING to FULL, Load

CSV ERP Architecture

GIU Nexus Schema Design

• Are

R1

Physical Config CLI Attributes

```
R1>enable
Password:
Password:
Password:
% Bad secrets

R1>enable
Password:
Password:
Password:
% Bad secrets

R1>enable
Password:
Password:
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#network 192.168.1.0 0.0.0.255 area 0
R1(config-router)#network 192.168.2.0 0.0.0.255 area 0
R1(config-router)#network 192.168.3.0 0.0.0.255 area 0
R1(config-router)#network 172.16.1.0 0.0.0.255 area 0
R1(config-router)#
00:50:47: %OSPF-5-ADJCHG: Process 1, Nbr 172.16.3.1 on Serial0/0/0 from LOADING to FULL, Loading Done

R1(config-router)#show ip ospf neighbor
^
% Invalid input detected at '^' marker.

R1(config-router)#exit
R1(config)#show ip ospf neighbor
^
% Invalid input detected at '^' marker.

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
172.16.3.1	0	FULL/ -	00:00:39	172.16.1.2	Serial0/0/0

R1#

Copy Paste

```
00:50:47: %OSPF-5-ADJCHG: Process 1, Nbr 172.16.3.1 on Serial0/0/0 from LOADING to FULL, Loading Done
```

```
R1(config-router)#show ip ospf neighbor
```

```
% Invalid input detected at '^' marker.
```

```
R1(config-router)#exit
```

```
R1(config)#show ip ospf neighbor
```

```
% Invalid input detected at '^' marker.
```

```
R1(config)#exit
```

```
R1#
```

```
%SYS-5-CONFIG_I: Configured from console by console
```

```
R1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
172.16.3.1	0	FULL/ -	00:00:39	172.16.1.2	Serial0/0/0

```
R1#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
```

```
* - candidate default, U - per-user static route, o - ODR
```

```
P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
```

```
C 172.16.1.0/24 is directly connected, Serial0/0/0
```

```
L 172.16.1.1/32 is directly connected, Serial0/0/0
```

```
O 172.16.2.0/24 [110/128] via 172.16.1.2, 00:01:51, Serial0/0/0
```

```
O 172.16.3.0/24 [110/65] via 172.16.1.2, 00:01:51, Serial0/0/0
```

```
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
```

```
C 192.168.1.0/24 is directly connected, GigabitEthernet0/0
```

```
L 192.168.1.1/32 is directly connected, GigabitEthernet0/0
```

```
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
```

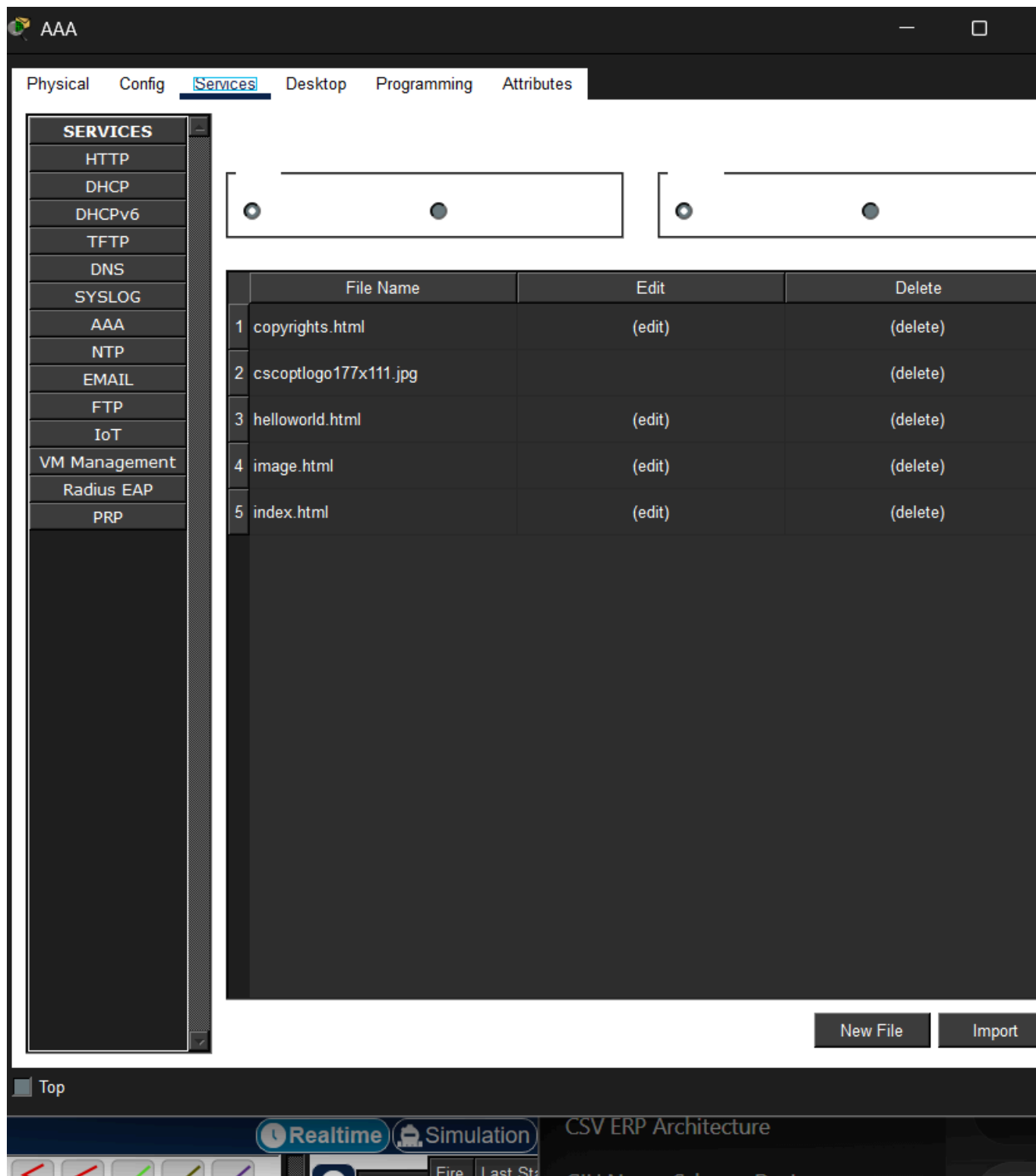
```
C 192.168.2.0/24 is directly connected, GigabitEthernet0/1
```

```
L 192.168.2.1/32 is directly connected, GigabitEthernet0/1
```

```
192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
```

```
--More--
```

[Copy](#)[Paste](#)[Top](#)



12. Part I – AAA Authentication on All Routers

This configures each router to query the AAA/TACACS+ server (192.168.1.2) for user authentication.

AAA on R1

```
R1(config)# aaa new-model
R1(config)# tacacs-server host 192.168.1.2 key TacacsKey123
```

```
R1(config)# aaa authentication login default tacacs+ local
R1(config)# aaa authentication enable default group tacacs+ enable
```

- `aaa new-model` enables the AAA framework.
- `tacacs-server host 192.168.1.2 key TacacsKey123` points R1 to the TACACS+ server with shared secret `TacacsKey123`.
- `aaa authentication login default group tacacs+ local` — First try TACACS+ for authentication; fall back to local database if TACACS+ is unavailable.
- `aaa authentication enable default group tacacs+ enable` — Use TACACS+ for `enable` mode authentication as well.

AAA on R2

```
R2(config)# aaa new-model
R2(config)# tacacs-server host 192.168.1.2 key TacacsKey123
R2(config)# aaa authentication login default group tacacs+ local
R2(config)# aaa authentication enable default group tacacs+ enable
```

AAA on R3

```
R3(config)# aaa new-model
R3(config)# tacacs-server host 192.168.1.2 key TacacsKey123
R3(config)# aaa authentication login default group tacacs+ local
R3(config)# aaa authentication enable default group tacacs+ enable
```

13. Part J – SYSLOG Logging on All Routers

This sends all router log messages to the SYSLOG server at `192.168.1.3`.

SYSLOG on R1

```
R1(config)# logging host 192.168.1.3
R1(config)# logging trap informational
R1(config)# logging on
```

SYSLOG on R2

```
R2(config)# logging host 192.168.1.3
R2(config)# logging trap informational
R2(config)# logging on
```

SYSLOG on R3

```
R3(config)# logging host 192.168.1.3
R3(config)# logging trap informational
R3(config)# logging on
```

`logging trap informational` sets the severity level — all messages at informational level and above (notice, warning, error, critical, alert, emergency) will be sent to the SYSLOG server.

14. Part K – Security Hardening (Passwords, SSH, Brute-Force)

Most of these steps were already completed during router initial configuration. This section confirms all requirements are met and fills in any remaining items.

K1 – Enable Password Hashing (Verify on All Routers)

All three routers already have:

- `enable secret cisco123Pa55` — hashed with MD5 (Type 5)
- `username ... secret ...` — all user passwords hashed
- `service password-encryption` — Type 7 encryption on remaining plain-text passwords

To verify, run `show running-config` and confirm all passwords appear as hashed strings (e.g., `$1$` prefix for MD5 or `7` for Type 7), never plain text.

K2 – Confirm SSH is Configured (Verify on All Routers)

On each router, verify:

```
R1# show ip ssh
```

Output should show: `SSH Enabled - version 2.0`

K3 – Confirm Brute-Force Protection (Verify on All Routers)

The command `login block-for 300 attempts 3 within 60` was already applied. Verify with:

```
R1# show login
```

K4 – Testing Console Access from Admin-Laptop

1. Double-click **Admin-Laptop** → **Desktop tab** → **Terminal**.
2. Default settings are fine (9600 baud, 8 data bits, no parity, 1 stop bit, no flow control). Click **OK**.
3. Press **Enter** — you should see: **R1 con0 is now available**
4. Login with: **Admin1** / **admin1Pa55**

15. Part L – IOS IPS Configuration on R1

IOS IPS (Intrusion Prevention System) is used to prevent employees from pinging R1. This is applied on R1's employee-facing interfaces (GigabitEthernet0/1 and GigabitEthernet0/2), in the **inbound** direction.

Note: Packet Tracer has limited IPS signature support. The configuration below follows the standard IOS IPS workflow. If your PT version does not support the signature-definition commands, skip to the "ACL Alternative" section at the end.

Step L1 – Create IPS Storage Directory on Flash

```
R1# mkdir flash:ipsdir
```

If prompted "Create directory filename [ipsdir]?" press **Enter**. If the directory already exists, that's fine.

```
R1# configure terminal
R1(config)# ip ips config location flash:ipsdir
```

Step L2 – Enable IPS Notification

```
R1(config)# ip ips notify log
```

Step L3 – Create IPS Rule

```
R1(config)# ip ips name IOSIPS
```

Step L4 – Configure Signature Categories

```
R1(config)# ip ips signature-category
R1(config-ips-category)# category all
R1(config-ips-category-action)# retired true
R1(config-ips-category-action)# exit
R1(config-ips-category)# category ios_ips basic
R1(config-ips-category-action)# retired false
```

```
R1(config-ips-category-action)# exit
R1(config-ips-category)# exit
```

This retires ALL signatures (to save resources) and then re-enables only the **ios_ips basic** category, which includes basic attack signatures.

Step L5 – Enable and Configure the ICMP Echo Signature (SID 2004)

```
R1(config)# ip ips signature-definition
R1(config-sigdef)# signature 2004 0
R1(config-sigdef-sig)# status
R1(config-sigdef-sig-status)# retired false
R1(config-sigdef-sig-status)# enabled true
R1(config-sigdef-sig-status)# exit
R1(config-sigdef-sig)# engine
R1(config-sigdef-sig-engine)# event-action deny-packet-inline
R1(config-sigdef-sig-engine)# exit
R1(config-sigdef-sig)# exit
R1(config-sigdef)# exit
```

Signature ID **2004** is the ICMP Echo Request (ping) signature. Setting **event-action deny-packet-inline** causes the router to silently drop the ICMP echo packets.

Step L6 – Apply IPS Rule to Inbound Interfaces

Apply to GigabitEthernet0/1 (Technical Support employees):

```
R1(config)# interface GigabitEthernet0/1
R1(config-if)# ip ips IOSIPS in
R1(config-if)# exit
```

Apply to GigabitEthernet0/2 (Employees' offices):

```
R1(config)# interface GigabitEthernet0/2
R1(config-if)# ip ips IOSIPS in
R1(config-if)# exit
```

The **in** direction means the IPS inspects all traffic entering R1 from those subnets.

16. Part M – Access Control Lists (ACLs) on R1

Two ACL requirements:

- **Requirement 8:** Permit only sales and marketing PCs to access the Web Server (HTTP, port 80).

- **Requirement 9:** Deny the Meeting Room PC from accessing the FTP Server (port 21).

Both requirements involve traffic originating from the employees subnet (192.168.3.0/24) and destined for the DMZ. We will create one combined **extended named ACL** and apply it on R1's **GigabitEthernet0/2** interface (employees-facing), in the **inbound** direction.

Step M1 – Create the Extended Named ACL

```
R1(config)# ip access-list extended EMPLOYEES_POLICY
```

Permit Sales PCs (192.168.3.2–.4) to access Web Server (172.16.3.2) on port 80:

```
R1(config-ext-nacl)# permit tcp host 192.168.3.2 host 172.16.3.2 eq 80
R1(config-ext-nacl)# permit tcp host 192.168.3.3 host 172.16.3.2 eq 80
R1(config-ext-nacl)# permit tcp host 192.168.3.4 host 172.16.3.2 eq 80
```

Permit Marketing PCs (192.168.3.5–.7) to access Web Server (172.16.3.2) on port 80:

```
R1(config-ext-nacl)# permit tcp host 192.168.3.5 host 172.16.3.2 eq 80
R1(config-ext-nacl)# permit tcp host 192.168.3.6 host 172.16.3.2 eq 80
R1(config-ext-nacl)# permit tcp host 192.168.3.7 host 172.16.3.2 eq 80
```

Deny all other hosts from accessing the Web Server on port 80:

```
R1(config-ext-nacl)# deny tcp any host 172.16.3.2 eq 80
```

Deny Meeting Room PC (192.168.3.8) from accessing FTP Server (172.16.3.3) on port 21:

```
R1(config-ext-nacl)# deny tcp host 192.168.3.8 host 172.16.3.3 eq 21
```

Permit all remaining IP traffic (so you don't accidentally block everything else):

```
R1(config-ext-nacl)# permit ip any any
R1(config-ext-nacl)# exit
```

Step M2 – Apply the ACL to GigabitEthernet0/2 Inbound

```
R1(config)# interface GigabitEthernet0/2
R1(config-if)# ip access-group EMPLOYEES_POLICY in
R1(config-if)# exit
```

ACL logic summary (top-to-bottom evaluation):

1. Sales-PC1 → Web Server port 80: **PERMITTED**
2. Sales-PC2 → Web Server port 80: **PERMITTED**
3. Sales-PC3 → Web Server port 80: **PERMITTED**
4. Mktg-PC1 → Web Server port 80: **PERMITTED**
5. Mktg-PC2 → Web Server port 80: **PERMITTED**
6. Mktg-PC3 → Web Server port 80: **PERMITTED**
7. Anyone else → Web Server port 80: **DENIED** (Meeting-PC, Mgmt-PC are blocked from web)
8. Meeting-PC → FTP Server port 21: **DENIED**
9. Everything else: **PERMITTED** (normal routing continues)

17. Part N – Save All Configurations

You **must** save configurations to NVRAM on every router so they survive a reload.

Save R1:

```
R1(config)# end
R1# copy running-config startup-config
```

When prompted "Destination filename [startup-config]?" press **Enter**.

Save R2:

```
R2(config)# end
R2# copy running-config startup-config
```

Press **Enter** when prompted.

Save R3:

```
R3(config)# end
R3# copy running-config startup-config
```

Press **Enter** when prompted.

Alternative shortcut command (same result): **R1# write memory** or **R1# wr**

18. Quick Verification Checklist

Use these commands and tests to verify everything is working:

Routing & Connectivity

Test	Command	Expected Result
Check OSPF neighbors on R1	<code>R1# show ip ospf neighbor</code>	R2 listed as FULL neighbor
Check OSPF neighbors on R2	<code>R2# show ip ospf neighbor</code>	R1 and R3 listed as FULL
Check routing table on R1	<code>R1# show ip route</code>	All subnets listed (O = OSPF learned)
Ping R2 from R1	<code>R1# ping 172.16.1.2</code>	Success (!!!!! = 5/5 packets)
Ping R3 from R1	<code>R1# ping 172.16.2.2</code>	Success
Ping Web Server from Sales-PC1	Desktop → Command Prompt → <code>ping 172.16.3.2</code>	Success

Security Verification

Test	Command/Action	Expected Result
Verify hashed passwords	<code>R1# show running-config</code>	All passwords appear as hash strings
Verify SSH is v2	<code>R1# show ip ssh</code>	SSH Enabled – version 2.0
Verify IPS on interfaces	<code>R1# show ip ips interfaces</code>	Gi0/1 and Gi0/2 show IOSIPS
Verify ACL exists	<code>R1# show access-lists</code>	EMPLOYEES_POLICY entries listed
Verify ACL applied on Gi0/2	<code>R1# show ip interface GigabitEthernet0/2</code>	Inbound access list is EMPLOYEES_POLICY
Test brute-force block	Try 3 wrong passwords on console	"Blocked" message appears
Ping R1 from IT-PC1	Command Prompt → <code>ping 192.168.2.1</code>	Request timed out (IPS blocks it)
Ping R1 from Sales-PC1	Command Prompt → <code>ping 192.168.3.1</code>	Request timed out (IPS blocks it)
HTTP from Sales-PC1 to Web	Desktop → Web Browser → <code>http://172.16.3.2</code>	Web page loads
HTTP from Meeting-PC to Web	Desktop → Web Browser → <code>http://172.16.3.2</code>	Request failed/blocked by ACL

Test	Command/Action	Expected Result
FTP from Meeting-PC to FTP	Desktop → Command Prompt → type ftp 172.16.3.3	Connection refused/blocked by ACL
Verify SYSLOG on server	Double-click SYSLOG-Server → Services → SYSLOG	Log entries appear from routers

Saving Confirmation

Command	Expected Result
R1# show startup-config	Config matches running-config
R2# show startup-config	Config matches running-config
R3# show startup-config	Config matches running-config

Complete Configuration Summary (Full CLI for Each Router)

R1 – Complete Running Configuration

```

hostname R1
service password-encryption
enable secret cisco123Pa55
!
username Admin1 privilege 15 secret admin1Pa55
username SSHAdmin1 privilege 15 secret SSHadmin1Pa55
!
ip domain-name company.local
ip ssh version 2
!
banner motd # WARNING: Unauthorized access is strictly prohibited! #
!
interface GigabitEthernet0/0
description Server_Room_LAN
ip address 192.168.1.1 255.255.255.0
ip ips IOSIPS in
no shutdown
!
interface GigabitEthernet0/1
description Tech_Support_LAN
ip address 192.168.2.1 255.255.255.0
ip ips IOSIPS in
no shutdown
!
interface GigabitEthernet0/2
description Employees_LAN
ip address 192.168.3.1 255.255.255.0
ip access-group EMPLOYEES_POLICY in
ip ips IOSIPS in
no shutdown

```

```
!  
interface Serial0/0/0  
  description WAN_Link_to_R2  
  ip address 172.16.1.1 255.255.255.0  
  clock rate 64000  
  no shutdown  
!  
router ospf 1  
  network 192.168.1.0 0.0.0.255 area 0  
  network 192.168.2.0 0.0.0.255 area 0  
  network 192.168.3.0 0.0.0.255 area 0  
  network 172.16.1.0 0.0.0.255 area 0  
!  
aaa new-model  
tacacs-server host 192.168.1.2 key TacacsKey123  
aaa authentication login default group tacacs+ local  
aaa authentication enable default group tacacs+ enable  
!  
logging host 192.168.1.3  
logging trap informational  
logging on  
!  
login block-for 300 attempts 3 within 60  
!  
ip ips config location flash:ipsdir  
ip ips notify log  
ip ips name IOSIPS  
!  
ip access-list extended EMPLOYEES_POLICY  
  permit tcp host 192.168.3.2 host 172.16.3.2 eq 80  
  permit tcp host 192.168.3.3 host 172.16.3.2 eq 80  
  permit tcp host 192.168.3.4 host 172.16.3.2 eq 80  
  permit tcp host 192.168.3.5 host 172.16.3.2 eq 80  
  permit tcp host 192.168.3.6 host 172.16.3.2 eq 80  
  permit tcp host 192.168.3.7 host 172.16.3.2 eq 80  
  deny    tcp any host 172.16.3.2 eq 80  
  deny    tcp host 192.168.3.8 host 172.16.3.3 eq 21  
  permit ip any any  
!  
line console 0  
  login local  
  exec-timeout 5 0  
line vty 0 4  
  login local  
  transport input ssh  
  exec-timeout 5 0  
!  
end
```

